REPORT ON

COWICHAN COPPER CO. LTD.
SUNRO MINE, JORDAN RIVER, B.C.

H.Hill & L.Starck & Associates Ltd. 11/7/63

REPORT ON

THE ECONOMICS

of

COWICHAN COPPER CO. LTD.

SUNRO MINE, JORDAN RIVER, B.C.

by: H. Hill & L. Starck & Associates Ltd.

July 19, 1963.

July 19, 1963.

REPORT ON

THE ECONOMICS

of

COWICHAN COPPER CO. LTD.

SUNRO MINE, JORDAN RIVER, B.C.

The following report is based on the ore reserves shown in the H. Hill & L. Starck & Associates Ltd. report of July 11, 1963, which is enclosed herewith. Operating costs used in this report are based on previous performance coupled with estimates for a future operation.

SUMMARY

Proven and indicated ore reserves at July 1963, are estimated at 1,695,000 tons, grading 1.57% copper after allowing for 5% dilution.

During the two year period, from July 1963 to June 1965, tonnage milled at the rate of 35,000 tons per month would total 825,000 tons grading 1.80% copper.

Ore reserves at July 1st, 1965, are estimated at 870,000 tons grading 1.33% copper providing no new ore is developed. Ore of this grade has a net smelter return value of $6.14 per ton after allowing for the reduced Consolidated Mining & Smelting Company royalty.

Operating profit during the two year period, after allowing for direct costs and the Consolidated Mining & Smelting Company royalty, but before allowing for capital expenditures or interest on loans, would amount to $2,722,550. Details are shown on Table I, which gives a forecast of operating profit for each month from July through December 1963, and for each quarter for the period January 1963 to July 1, 1965.

To maintain the costs used in the forecast will require maximum effort by all personnel. A close analysis should be made of the performance of all supervisors with a view to obtaining the greatest possible efficiency.

Summary (Continued)

The operating costs are based on the assumption that adequate finances will be available so that management can make the planned, necessary capital improvements when required.

TONS MILLED

The increase in tons milled to 35,000 per month is based on improvements in the crushing plant to provide 1,500 tons per day of minus 5/8" ore to the rod mills.

MILL HEADS

Estimated mill heads are based on the forecast for ore removal shown in Table 2 and Maps 1, 2 and 3.

Grades shown are based on the ore reserve report by H. Hill & L. Starck & Associates Ltd., dated July 11, 1963, after allowing for dilution by 5% of material assaying 0.1% copper.

NET SMELTER RETURN

The net smelter return per pound of copper in the mill heads was calculated as follows:

Assays: per metric ton (2204.6 lbs.) of concentrate

Copper - 25.12% (average of concentrates produced to June 21, 1963)
Gold - 4.25 gms. per metric ton
Silver - 42.5 gms. per metric ton

Quotations: Month of June 1963

Copper - E.M.J. export refinery quotation 28.39¢ per lb.
Gold - U.S. Treasury net price quotation $34.9125 per oz.
Silver - Handy & Harman, N.Y. quotation for foreign fine silver $1.2768 per oz. less discount of 0.004 = $1.2728 per oz.

Contents & Value:

Contents	Contents Paid For	Price	Value
553.80 lb. copper	531.75 lbs.	28.396¢-1	$145.68 U.S. funds
4.25 gms. or 0.137 oz. Au.	.137 x .95	$34.9125	4.54 U.S. funds
42.5 gms. or 1.37 oz. Ag.	1.37 x .90	$ 1.2728	1.56 U.S. funds
		Total	$151.78 U.S. funds
Freight and treatment - $18.26 per dry metric ton			18.25
			$133.52 U.S. funds

Net Smelter Return

Contents & Value (Continued)

Total carried forward from page 2	\$133.53 U.S. funds
Using June 25th exchange rate of 1.0775	\$143.88 Canadian funds
Hauling-mine to dock and loading on board ship	2.05
	\$141.83 Canadian funds

Net smelter return per lb. copper in concentrate = $\frac{141.83}{553.8}$ = 25.61¢

Net smelter return per lb. copper in mill heads using a mill recovery of 95% = 25.61 x .95 = 24.33¢

OPERATING COSTS

(a) Mining

During July, August and September 1963, most of the production will come from the broken ore in the B North stope. During October, November and December 50% of production will come from the narrower C ore body. After January 1, 1964, 50% to 55% of production will be hoisted from below the 5130 level and the remainder will come from the C ore body.

Allowance has been made in the mining cost estimate for the increased cost of mining the C ore body and increased wages and supply costs.

(b) Development and Exploration

This item includes stope preparation, development and diamond drilling to outline the ore bodies, and exploratory diamond drilling. For the River Zone, the cost of already completed development plus an estimate of that still required to complete development of the zone is \$0.60 per ton. This is the figure at which development has been written off over tonnage milled to date, resulting in the deferred development item in the April 30, 1963 balance sheet. In view of the fact that development of the narrower ore bodies, remaining after the River B ore body has been extracted, will cost more per ton, a figure of \$1.00 per ton has been used in Table I. Allowance has been made to maintain a workable broken ore reserve ahead of production.

(c) Milling

The reduction of this figure to \$1.28 per ton from the \$2.00 per ton which milling has cost to date, is based on the improvements in the crushing plant which will provide 1500 tons per day of minus 5/8" ore to the rod, and on other improvements in the mill, if found necessary, to maintain a monthly tonnage of 35,000.

ADMINISTRATION

During the fiscal year ending April 30, 1963, the cost of administration was $0.72 per ton milled for an average of 19,000 tons per month. The figure of 0.45 per ton, used in Table I, is this cost spread over 35,000 tons per month.

ROYALTY

The current royalty arrangement with Consolidated Mining & Smelting Company per ton of ore milled is based on the following formula:

(.90 x grade x export refinery price ÷ 30) converted to Canadian funds.

It ends with the milling of 900,000 tons of 1.8% ore. Remaining ore is subject to a royalty per ton calculated by the following formula:

(.25 x grade x export refinery price ÷ 30) converted to Canadian funds.

The change occurs in the period October to December 1964 in Table I.

CAPITAL EXPENDITURES

The figures shown in Table I allow for completion of the shaft installation, new crusher and installation and miscellaneous expenditures.

H. HILL & L. STARCK & ASSOCIATES LTD.

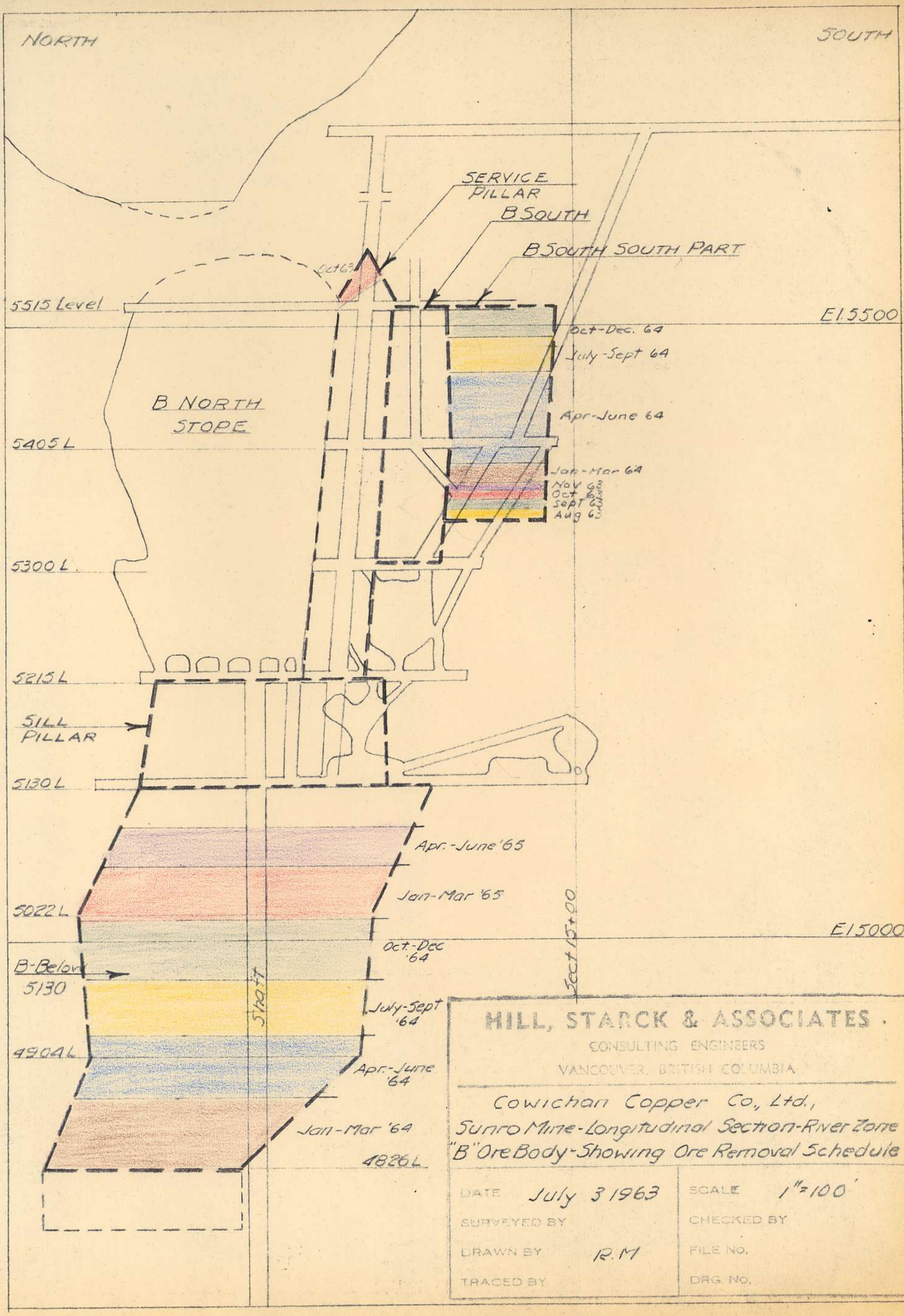
H. Hill

Henry L. Hill

L. Starck

L. P. Starck

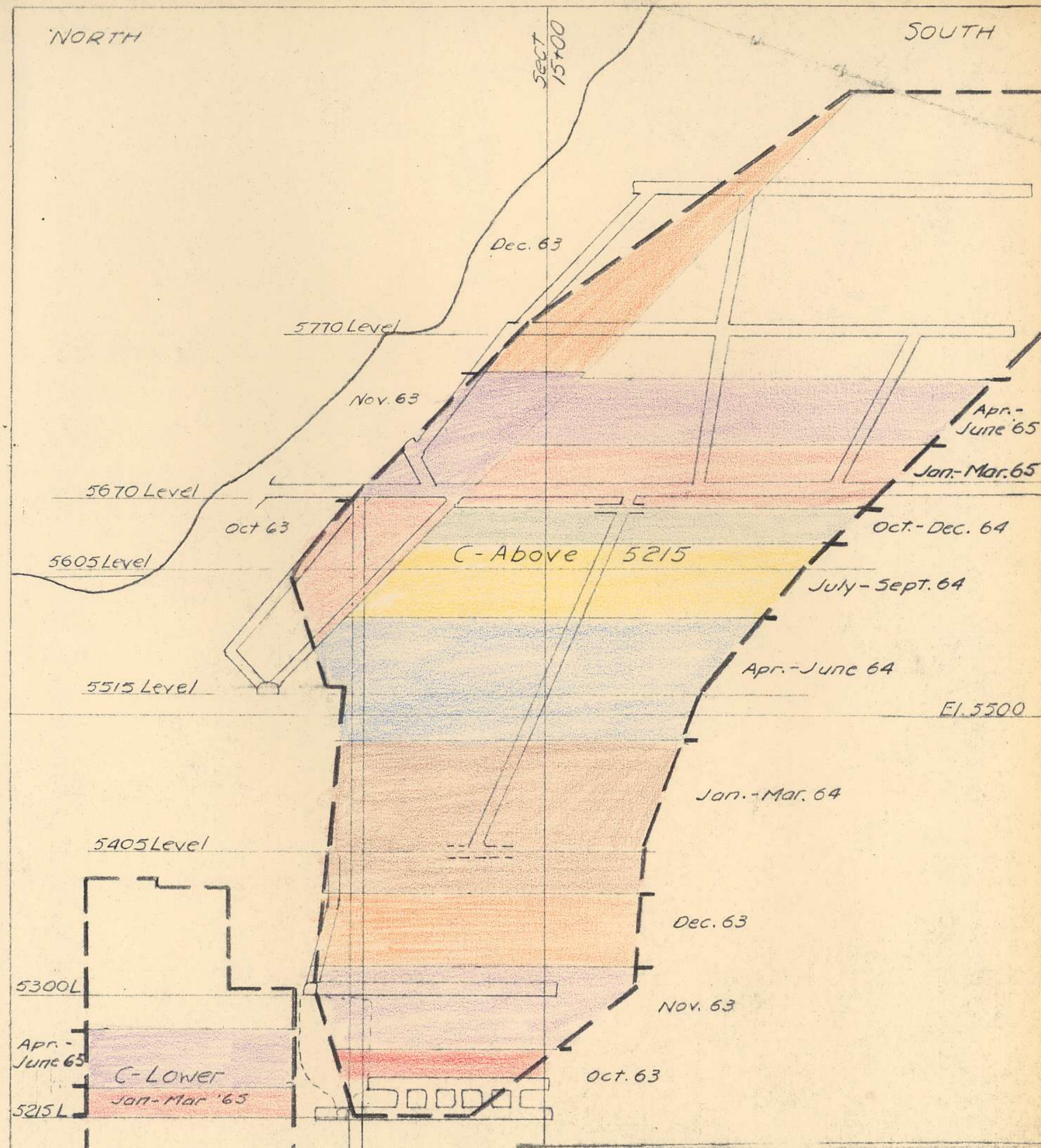
HLH:LPS/mjr



HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.,
Sunro Mine - Longitudinal Section - River Zone
"B" Ore Body - Showing Ore Removal Schedule

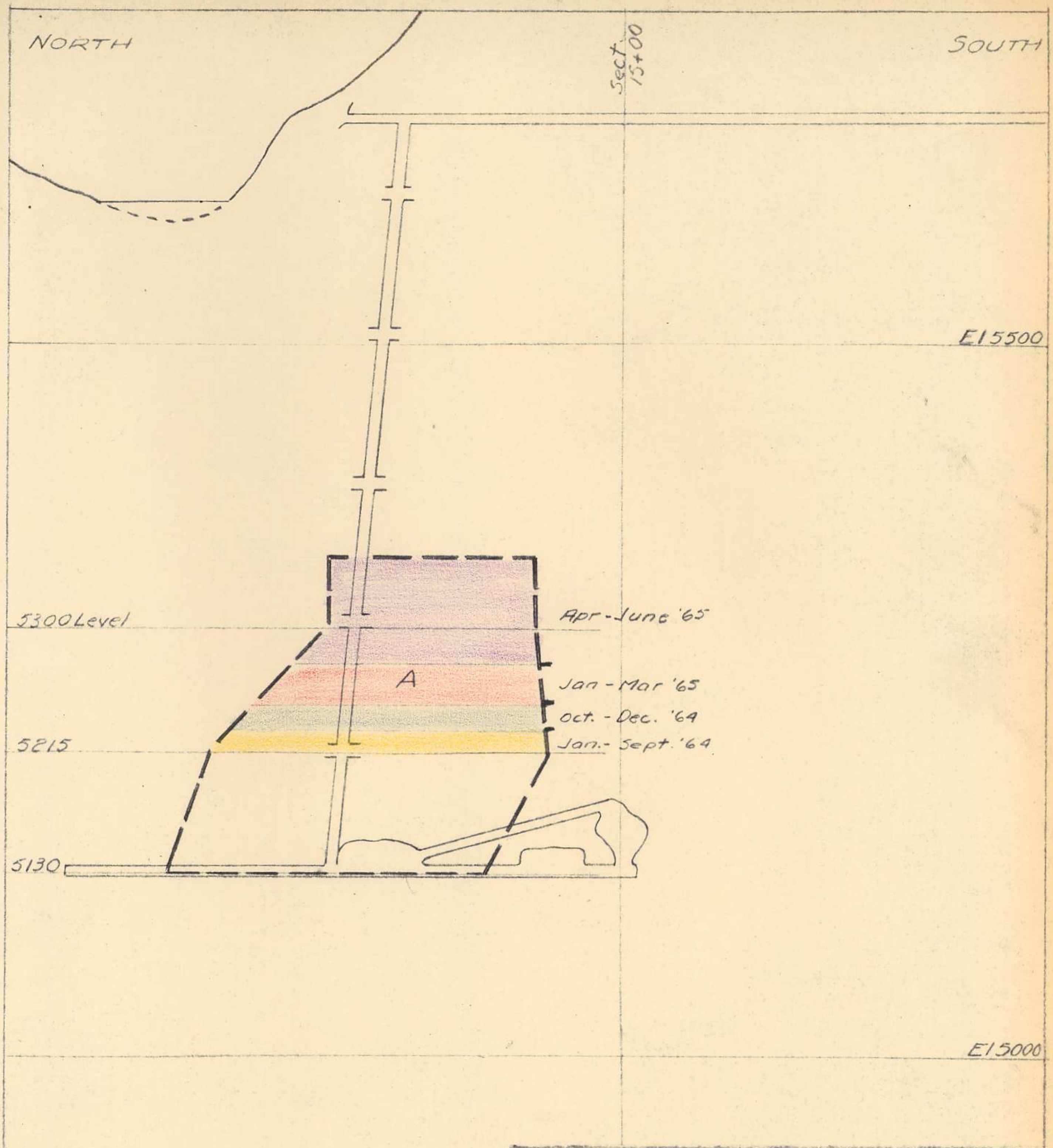
DATE	July 3 1963	SCALE	1" = 100'
SURVEYED BY		CHECKED BY	
DRAWN BY	R.M	FILE NO.	
TRACED BY		DRG. NO.	



HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co, Ltd.,
Sunro Mine - Longitudinal Section - River Zone
C Ore Body Showing Ore Removal Schedule

DATE	July 3 1963	SCALE	1" = 100'
SURVEYED BY		CHECKED BY	
DRAWN BY	R.M.	FILE NO.	
TRACED BY		DRG. NO.	



HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co, Ltd.,
Sunro Mine-Longitudinal Section-River Zone
"A" Ore Body-Showing Ore Removal Schedule

DATE	July 3 1963	SCALE	1"=100'
SURVEYED BY		CHECKED BY	
DRAWN BY	R.M.	FILE NO.	
TRACED BY		DRG. NO.	

Period	July 1963		August 1963		September 1963		October 1963		November 1963		December 1963		January - March 1964		April - June 1964		July - September '64		October - December '64		January - March '65		April - June '65		Total
Tons Milled	25 000		30 000		35 000		35000		35000		35 000		105000		105000		105000		105000		105000		105000		825000
Mill Heads Per Cent Copper	1.86		1.92		1.91		1.80		1.95		1.91		1.88		1.70		1.76		1.81		1.79		1.66		1.80
	Per Ton	Per Month	Per Ton	Per Month	Per Ton	Per Month	Per Ton	Per Month	Per Ton	Per Month	Per Ton	Per Month	Per Ton	Per Period	Per Ton	Per Period	Per Ton	Per Period	Per Ton	Per Period	Per Ton	Per Period	Per Ton	Per Period	
Net Smelter Return	$9.05	$226250	$9.34	$280200	$9.29	$325150	$8.76	$306600	$9.49	$332150	$9.29	$325150	$9.15	$960750	$8.27	$868350	$8.56	$898800	$8.81	$925050	$8.71	$914550	$8.08	$848400	
Operating Costs Mining	$0.85		0.85		0.90		1.00		1.20		1.30		1.45		1.45		1.45		1.45		1.45		1.45		
Development & Explor.	0.85		1.00		1.00		1.00		1.00		1.00		1.03		1.03		1.03		1.03		1.03		1.03		
Milling	1.55		1.40		1.25		1.25		1.25		1.25		1.28		1.28		1.28		1.28		1.28		1.28		
Administration	0.63		0.52		0.45		0.45		0.45		0.45		0.45		0.45		0.45		0.45		0.45		0.45		
Total	3.88		3.77		3.60		3.70		3.90		4.00		4.21		4.21		4.21		4.21		4.21		4.21		
Operating Profit	5.17		5.57		5.69		5.06		5.59		5.29		4.94		4.06		4.35		4.60		4.50		3.87		
Royalty (C.M.&S)	1.71		1.76		1.75		1.65		1.79		1.75		1.73		1.56		1.62		1.41		0.46		0.42		
Operating Profit less C.M.&S. Royalty	3.46	86500	3.81	114300	3.94	137900	3.41	119 350	3.80	133 000	3.54	123900	3.21	337050	2.50	262500	2.73	286650	3.19	334950	4.04	424200	3.45	362250	
Cumulative Operating Profit less Royalty		86500		200800		338 700		458050		591050		714950		1052000		1314500		1601150		1936100		2360300		2722550	
Capital Expenditure $240,000		20000		20000		20000		20000		10 000		10000		30000		30000		30000		16700		16700		16600	

Note –
Proven & Indicated ore reserves as at July 1st 1965 estimated at 870,000 tons @ 1.33% Cu.
Net smelter returns after allowing for reduced royalty = $6.14 at present metal prices & exchange

TABLE I

NOTE – The above forecast is based on the following:

(1) Suitable improvement in the crushing plant to guarantee 1500 tons per day of minus 5/8" ore to the Rod Mills to maintain minimum of 35,000 T/Month.

(2) Close supervision of all operations

H. Hill & L. Starck & Associates Ltd
July 10 1963

6 off each

Cowichan Copper Co., Ltd. - Sunro Mine

Ore Schedule   Note: Grades shown except * are after dilution by 5% of 0.1% Cu.

Period	July 1963		August 1963		Sept. 1963		October 1963		November '63		December '63		Jan.-Mar '64		April-June '64		July-Sept '64		Oct-Dec '64		Jan-Mar '65		Apr-June '65		Total	
	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	%Cu	Tons	
B - North Stope	20200	1.90*	25500	1.90*	30500	1.90*	7000	1.90*	6300	1.90*															89500	
B - South																										
B - ½ Service Pillar above 5515 North End C - above 5515							13000	1.52	8000	1.87	3770 5510	1.88 1.36													30280	
B - South - South Part	300	3.05	1300	3.05	1300	3.05	1300	3.05	1300	3.05			2600	3.05	6500	3.05	3900	3.05	2600	3.05					21100	
B - Service Pillar																										
B - Sill Pillar																										
B - 4800 to 5130							3900	2.02	3900	2.02	3900	2.02	60400	2.02	57500	2.02	57500	2.02	53600	2.02	47800	2.02	20700	2.02	309200	
B - Below 4800																					3900	1.91	13900	1.91	17800	
C - Above 5670	2600	1.59	2000	1.59	2000	1.59	2000	1.59													25525	1.77	30000	1.77	64125	
C - 5230 - 5670	1900	1.55	1200	1.62	1200	1.62	7800	1.85	15500	2.04	21820	2.04	16880 21220	2.04 1.34	7380 29720	1.34 0.95	18880 20820	0.95 1.67	25400	1.67	8275	1.67			197995	
Lower C																			7800	1.77	3900	1.77	10000	1.77	21700	
Cross Fault																										
"A" Zone above 5215													3900	1.15	3900	1.15	3900	1.15	15600	1.15	15600	1.15	30400	1.15	73300	
TOTAL	25000	1.86	30000	1.92	35000	1.91	35000	1.80	35000	1.95	35000	1.91	105000	1.88	105000	1.70	105000	1.76	105000	1.81	105000	1.79	105000	1.66	825000	1.80

TABLE 2

H. Hill & L. Starck & Associates Ltd.   July 6, 63

REPORT

on

COWICHAN COPPER CO. LTD.

SUNRO MINE, JORDAN RIVER, B.C.

ORE RESERVES AS AT JUNE 15, 1963

by: H. Hill & L. Starck & Associates Ltd.

July 11, 1963

July 11, 1963.

REPORT

on

COWICHAN COPPER CO., LTD.

SUNRO MINE, JORDAN RIVER, B.C.

ORE RESERVES AS AT JUNE 15, 1963.

The following report is based on a study of the company maps and records during the week of June 16th to June 22nd, 1963, and an examination of the underground workings by the writers.

SUMMARY

Ore reserves as at June 15th, 1963 are summarized below:

	Broken Ore		Proven Ore		Indicated Ore		Total	
	Tons	% Cu	Tons	% Cu	Tons	% Cu	Tons	% Cu
River Zone	102,500	1.89	1,019,600	1.79	59,100	1.93	1,181,200	1.81
Cave Zone					340,900	1.23	340,900	1.23
Center Zone					38,300	1.37	38,300	1.37
New Zone					54,600	1.20	54,600	1.20
Total	102,500	1.89	1,019,600	1.79	492,900	1.32	1,615,000	1.65

No allowance for dilution has been made in these figures

GEOLOGY

The geology of the Sunro Mine area is described in the 1950 Annual Report of the B. C. Minister of Mines, pages 180-193.

The ore bodies occur in shear zones in the Metchosin volcanics near a sill like intrusion of gabbro. The Metchosin volcanics, which are of Tertiary (Eocene) age, are predominately basalt in the mine area. The basalt is a fine grained, dark greenish grey rock consisting of plagioclase and dark green hornblende. Near the mineralized zones the plagioclase has been largely altered to hornblende.

Geology (Continued)

The gabbro is a coarse grained, dark greenish grey rock of Tertiary (Oligocene) age, with conspicuous plagioclase crystals. The ferro-magnesian mineral in the gabbro, near the mineralized zones, is principally hornblende with some augite.

About one and a half miles to the north of the mine workings the Leech River overthrust fault outcrops and separates the Tertiary rocks of the mine area from the overthrust, pre-Tertiary, metamorphic rocks to the north. The fault strikes east and dips 45° to 75° to the north. Northwest trending shear zones in the Metchosin volcanics, including those in which the ore bodies are found, may be related to the Leech River fault.

The sulphides in the ore zones are, in order of abundance, chalcopyrite, pyrrhotite, pyrite, and small amounts of molybdenite. Very small amounts of pentlandite have been noted in the pyrrhotite. The ore sulphides form a pattern of gash-like veinlets and lenticular masses in the hornblendized rock of the shear zones. A small amount of chalcopyrite occurs as disseminated grains.

Twelve zones of mineralization have been recognized. The River, Cave, Center and New zones have received the most attention, and contain the ore reserves shown in this report.

The River zone, which is currently being developed and mined, is in basalt, and strikes N 20° W with an almost vertical dip. It is characterized by numerous, steeply dipping slips, which converge at the northwest end, and form an ore body up to 100 feet wide. To the southeast the slips diverge with ore lenses along the stronger slips, and persist for several hundred feet before the mineralization dies out. Four ore bodies are recognized in this zone:

(1) The B ore body, which includes the wide northwest end of the zone and has been developed between elevations 4800 and 5550;

(2) The C ore body, which diverges to the southeast of B, and has been developed between elevations 5200 and 5950;

(3) The Lower C ore body, which lies 100 feet east of the B, between elevations 5000 and 5400;

(4) The A ore body, which diverges to the southwest from the B, and lies between elevations 5130 and 5350. Below elevation 5130 this ore body merges with the B.

The River zone has been developed to date over a length of 800 feet and a depth of 1150 feet.

Geology (Continued)

The Cave zone has been roughly outlined by the Cave adit and by limited diamond drilling. It lies 600 feet to the southwest of the River zone between elevations 5080 and 5550, strikes N 40° W, and dips vertically. It is in hornblendized basalt at the contact with a body of gabbro.

The Center zone has been indicated between elevations 5250 and 5625 by the Center adit, and two drill holes. It lies some 400 feet west of the River zone on the north side of the Jordan River, is in basalt, strikes N 15° W and dips steeply to the east.

The New zone has been indicated by seven drill holes over a length of 300 feet between elevations 5080 and 5550. It lies on the north side of the Jordan River, 100 feet northeast of the Center zone, and is in basalt. The strike indicated is N 30° W and the dip is steep to the northeast.

PROVEN AND INDICATED ORE RESERVES - Table I

General

The ore reserve calculations are based, for the most part, on diamond drill hole intersections. In the River zone these are flat holes drilled on approximately 50 foot spacing from levels about 100 feet apart. Where they exist, channel samples and up and down holes intersecting the ore body between levels have also been used in the calculations.

In the Cave, Center and New zones the indicated reserves are based on more widely spaced holes, and on channel samples taken in the Cave and Center adits.

Assays used and calculations are shown on Maps 1 to 20 inclusive. Map 1 is an index plan showing the location of the ore blocks in plan. Maps 21, 22 and 23 are longitudinal sections showing the location of the ore blocks in elevation.

River Zone

B North Stope: This block, together with part of B South, were the first to be mined. Current production is almost entirely from here, and the tonnage of broken ore shown is calculated from stope volume. The grade is calculated from mill heads during the past several months, adjusted for the small tonnage drawn from other working places.

B South: The broken ore tonnage shown is for the part of the block already mined between the 5215 and 5300 levels. The grade for the broken ore is that calculated for the unmined portion of the block.

Proven and Indicated Ore Reserves (Continued)

The proven ore in place for this block lies between the 5300 and the 5515 level. Calculations are shown on Maps 5, 6, 7 and 17, and are based on diamond drill holes on the 5515, 5405 and 5300 levels.

B South - South End: This is a narrow high grade body lying between elevations 5540 and 5338, marking the south end of the B ore body. Calculations are shown on Maps 5, 6 and 18.

B Service Pillar: This block contains the main service raise for the B and C workings, and lies between the B North and B South stopes, and between elevations 5561 and 5215. Calculations are shown on Maps 5, 6, 7, 8 and 17, and are based on drill holes on the 5515, 5405, 5300 and 5215 levels. out

B Sill Pillar: This block lies between the 5215 scram level and the 5130 haulage level, and below the B North Stope and B Service Pillar blocks. Calculations are shown on Maps 9, 10 and 14 , and are based on drill holes and channel samples on the 5130 and 5215 levels, and down holes just below the 5130 level.

B Ore Below 5130: This ore lies below the main haulage level, and will be serviced by the shaft now nearing completion. Calculations are shown on Maps 9 to 12 inclusive and 16, and are based on flat holes on the 5130, 5022, 4904 and 4800 levels.

C - Above 5215: This ore lies between the 5215 and 5940 levels and is on the comparatively narrow southeast split of the B ore body. Calculations are based on up holes above the 5770 level, on flat holes on the 5770, 5670, 5605, 5515, 5405, 5300 and 5215 levels, and on holes between the 5500 and 5605 levels, the 5405 and 5515 levels, and the 5300 and 5405 levels, and on channel samples on the 5670 level. Calculations are shown on Maps 2 to 7 inclusive, and Maps 15 and 16.

C Lower: This ore body lies to the east of the B ore body, and is an eastward dipping split of the B ore body between elevations 5000 and 5485. Calculations are based on down holes between the 5300 and 5405 levels, between the 5305 and 5130 levels and below the 5130 level, and on flat holes on the 5305 and 5130 levels. The southeast end of this ore body passes through the mill excavation. Calculations are shown on Maps 7 to 10 inclusive and on 13.

A Ore Body: This is a southwest split from the B ore body, diverging from it above the 5022 level, and carries through to the 5350 level. A zone of low grade material and three high grade stringers occupy the position of the A ore body on the 5405 and 5515 levels respectively. Calculations are shown on Maps 7, 8, 9 and 18.

Proven and Indicated Ore Reserves (Continued)

Cross Fault: This ore body shows in drill holes on the 5130 level. There is some indication from drilling below this level that it exists there also and occupies a shear zone almost at right angles to the B ore body. Calculations are shown on Maps 9 and 18.

Cave, Center and New Zones: These ore zones lie to the southwest of the River zone, and are indicated by limited drilling from the surface and the 5130 adit, and by channel samples in the Cave and Center adits. Calculations are shown on Maps 19 and 20.

H. HILL & L. STARCK & ASSOCIATES LTD.

R. P. Mason

R. P. Mason

H Hill

Henry L. Hill

RPM:HLH/mjr

	BLOCK	BROKEN ORE			PROVEN ORE			BROKEN + PROVEN			INDICATED ORE			TOTAL		
		Tons	%Cu	T x %	Tons	%Cu	T x %	Tons	%Cu	T x %	Tons	%Cu	T x %	Tons	%Cu.	T x %
	River Zone															
caved X	B North Stope	100500	1.90	190950				100,500	1.90	190950				100 500	1.90	190950
Pillar X	B South	2000	1.20	2400	48600	1.20	58320	50.600	1.20	60720	-			50600	1.20	60720
Doubtful X	B South South End				20100	3.20	64320	20,100	3.20	64320				20100	3.20	64320
100' Pillar X	B Service Pillar				108800	1.62	176256	108 800	1.62	176256				108800	1.62	176256
100' Sill X	B Sill Pillar				37300	2.24	83552	37 300	2.24	83552				37300	2.24	83552
Part	B Ore below 5130				303500	2.12	643320	303500	2.12	643.320	35100	2.00	70200	338600	2.11	713520
✓	C Above 5215				339000	1.64	555960	339000	1.64	555960				339000	1.64	555960
?	C Lower				56600	1.85	104710	56600	1.85	104710				56600	1.85	104710
most in pillar ?	A Ore Body				105700	1.31	138467	105700	1.31	138467				105 700	1.31	138467
?	Cross Fault										24000	1.82	43680	24000	1.82	43680
	Total - River Zone	102500	1.89	193350	1,019,600	1.79	1824905	1,122,100	1.80	2018255	59 100	1.93	113880	1,181,200	1.81	2132135
	Cave Zone										340,900	1.23	419307	340,900	1.23	419307
	Centre Zone										38,300	1.37	52471	38,300	1.37	52471
	New Zone										54,600	1.20	65520	54,600	1.20	65520
	Total	102 500	1.89	193350	1019600	1.79	1824905	1,122100	1.80	2018255	492900	1.32	651178	1,615,000	1.65	2669433

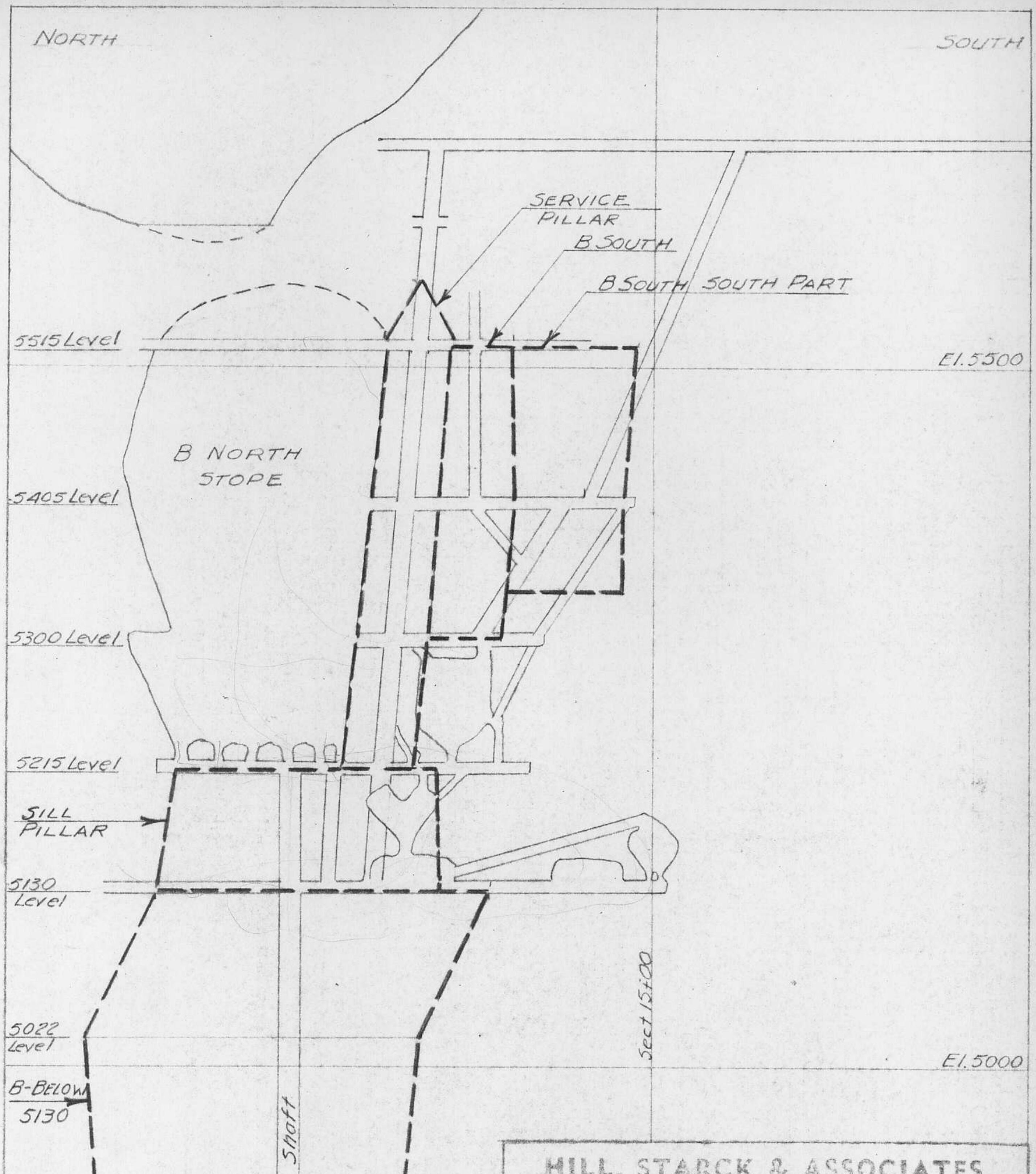
Note :- No Allowance made for dilution.

TABLE 1.

HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co, Ltd.,
Sunro Mine – Summary of Proven and Indicated Ore Reserves.

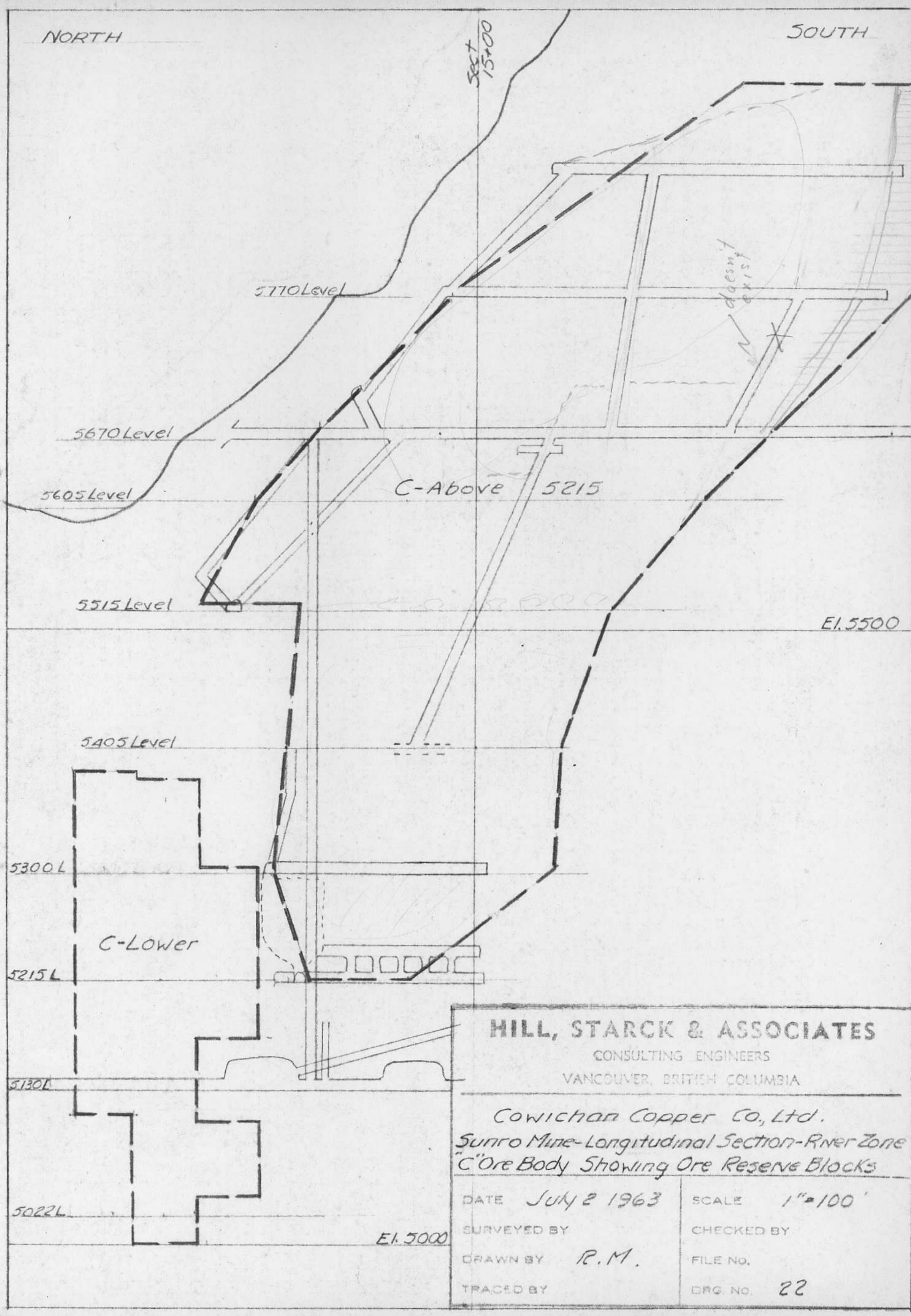
DATE June 28 '63	SCALE –
SURVEYED BY	CHECKED BY
DRAWN BY R.M.	FILE No.
TRACED BY	DRG. No.



HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.,
Sunro Mine-Longitudinal Section-River Zone
B Ore Body Showing Ore Reserve Blocks.

DATE July 2 1963	SCALE 1" = 100'
SURVEYED BY	CHECKED BY
DRAWN BY R.M.	FILE NO.
TRACED BY	DRG. NO. 21



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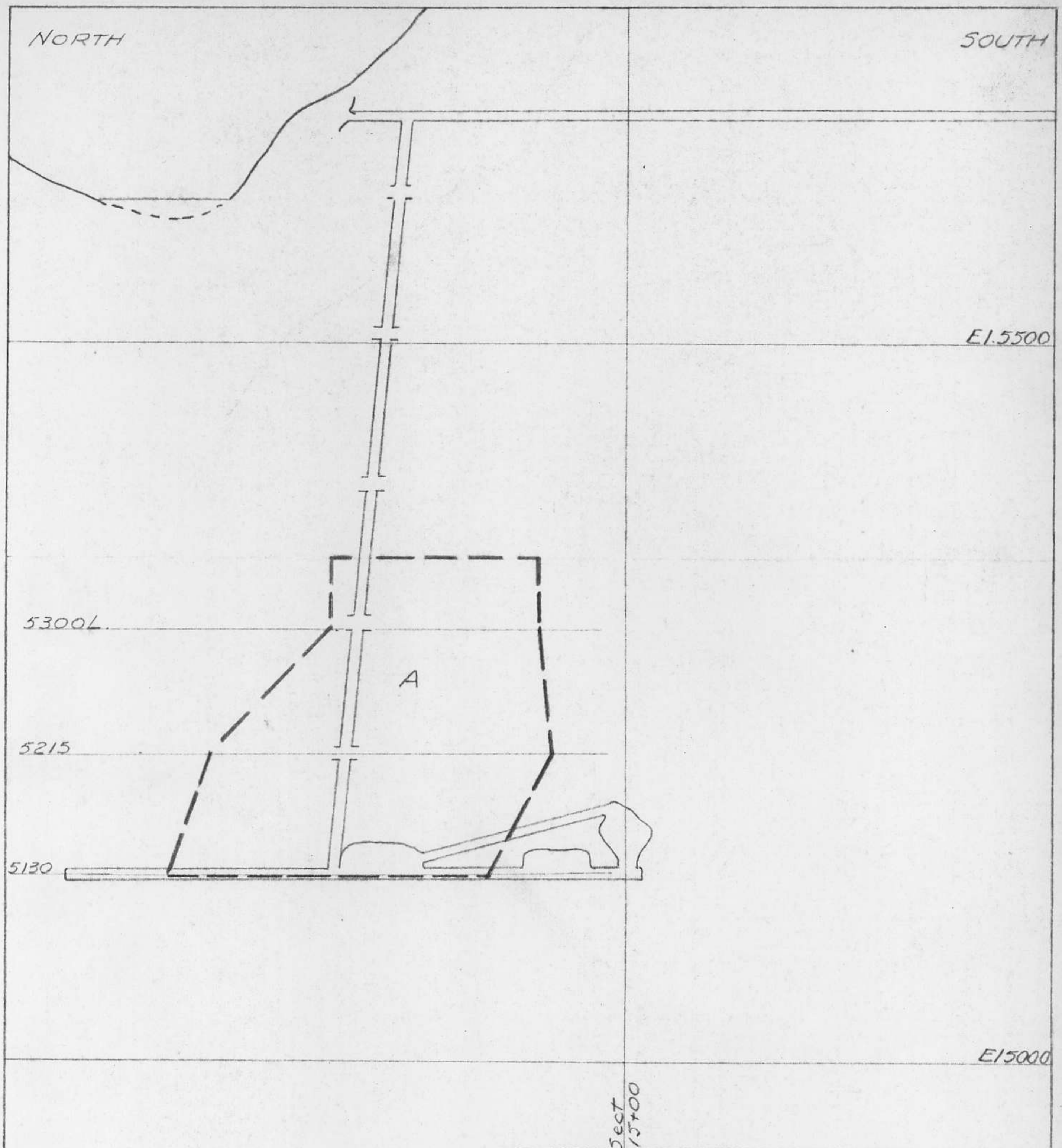
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.

Sunro Mine - Longitudinal Section - River Zone

"C" Ore Body Showing Ore Reserve Blocks

DATE July 2 1963	SCALE 1" = 100'
SURVEYED BY	CHECKED BY
DRAWN BY R.M.	FILE NO.
TRACED BY	DRG. NO. 22



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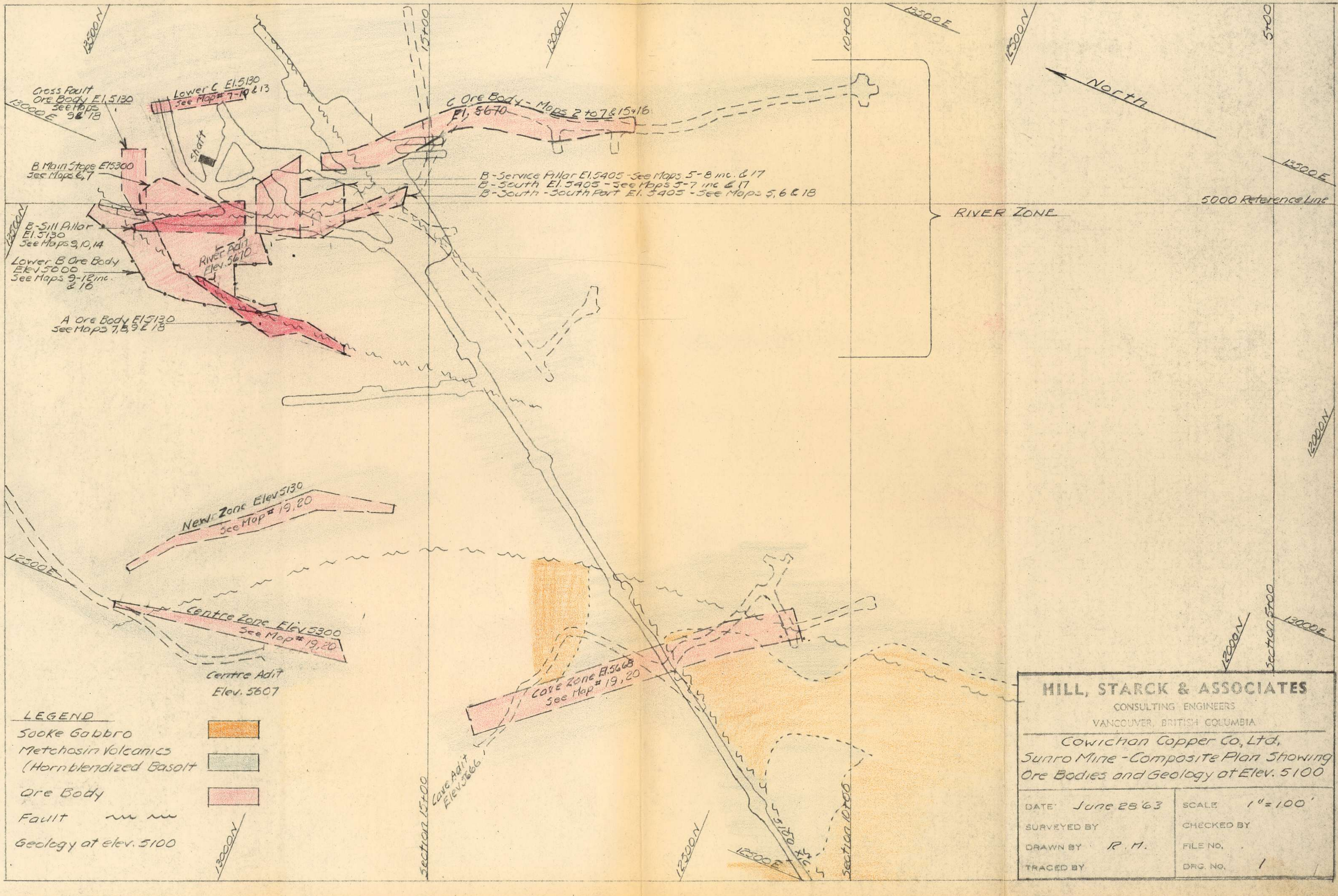
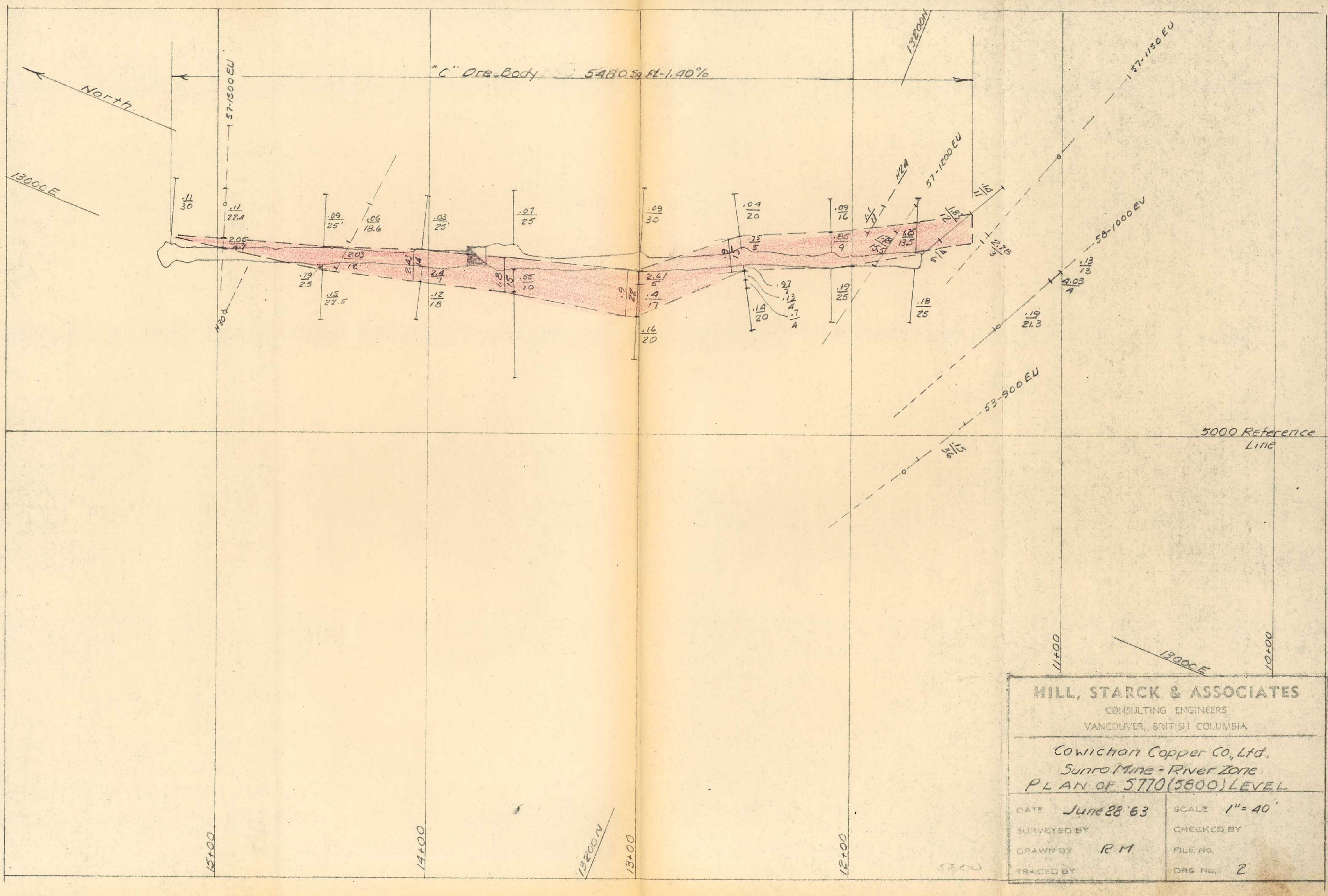
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.
Sunro Mine-Longitudinal Section-River Zone
"A" Ore Body-Showing Ore Reserve Blocks

DATE July 3 1963	SCALE 1"=100'
SURVEYED BY	CHECKED BY
DRAWN BY R.M.	FILE NO.
TRACED BY	DRG. NO. 23



13500N
15+00
13000N
10+00
13500E
12500N
5+00
North
Cross Fault Ore Body El. 5130 See Maps 9 & 18
13000E
Lower C El. 5130 See Map # 7-10 & 13
C Ore Body - Maps 2 to 7 & 15 & 16 El. 5670
Shaft
B Main Stope El. 5300 See Maps 6, 7
B - Service Pillar El. 5405 - See Maps 5 - 8 inc. & 17
B - South El. 5405 - See Maps 5 - 7 inc & 17
B - South - South Part El. 5405 - See Maps 5, 6 & 18
13500E
5000 Reference Line
RIVER ZONE
B - Sill Pillar El. 5130 See Maps 9, 10, 14
River Adit Elev. 5670
Lower B Ore Body Elev. 5000 See Maps 9-12 inc. & 16
A Ore Body El. 5130 See Maps 7, 8, 9 & 18
12000N
New Zone Elev 5130 See Map # 19, 20
12500E
Centre Zone Elev 5300 See Map # 19, 20
Centre Adit Elev. 5607
Cove Zone El. 5668 See Map # 19, 20
Cove Adit Elev. 5666
12000N
Section 5+00
13000E
LEGEND
Sooke Gabbro
Metchosin Volcanics (Hornblendized Basalt
Ore Body
Fault
Geology at elev. 5100
13000N
Section 15+00
12500N
12500E
Section 10+00
HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co, Ltd,
Sunro Mine - Composite Plan Showing Ore Bodies and Geology at Elev. 5100
DATE: June 28 '63
SCALE 1" = 100'
SURVEYED BY
CHECKED BY
DRAWN BY R. M.
FILE NO.
TRACED BY
DRG. NO. 1



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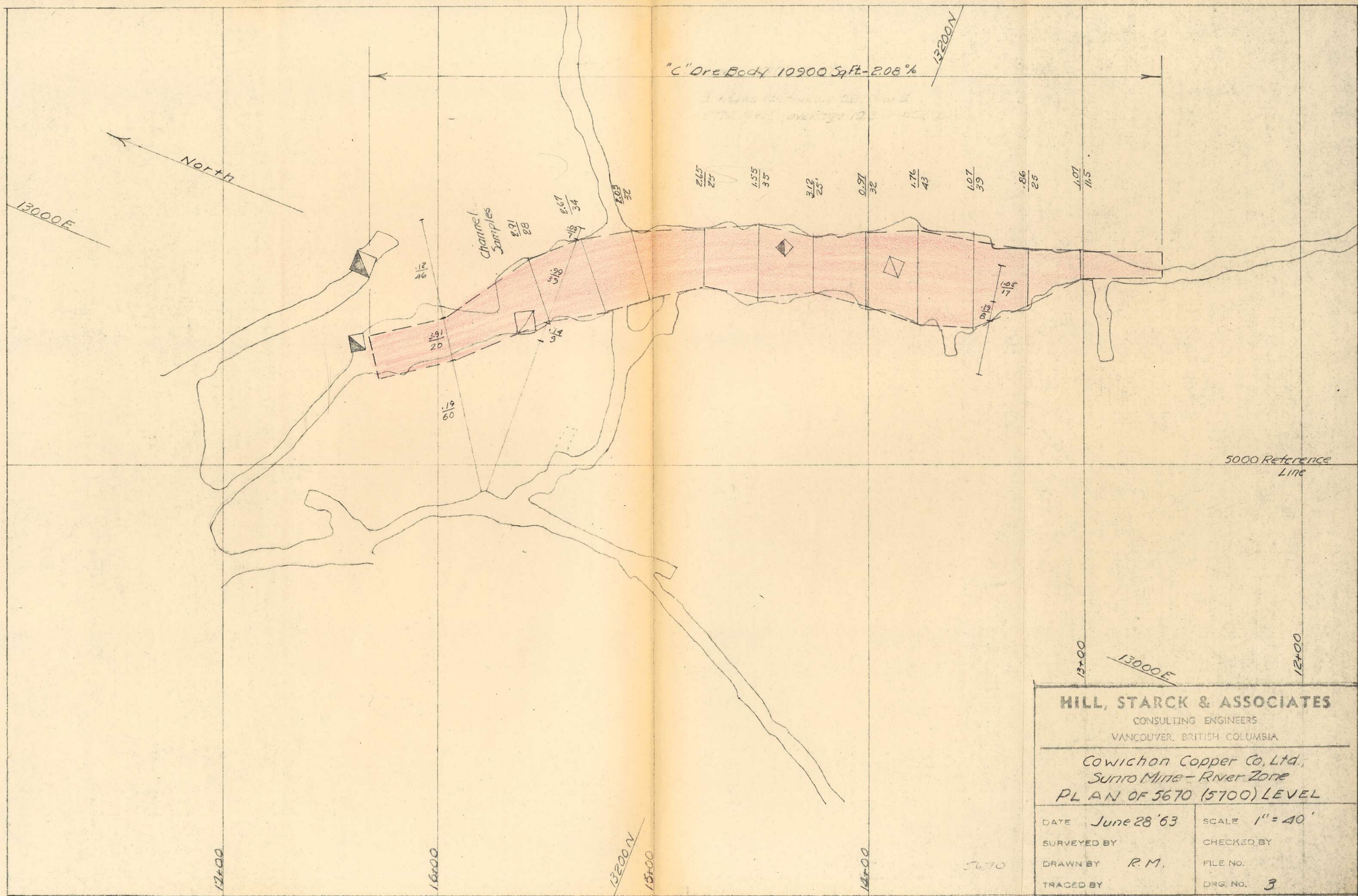
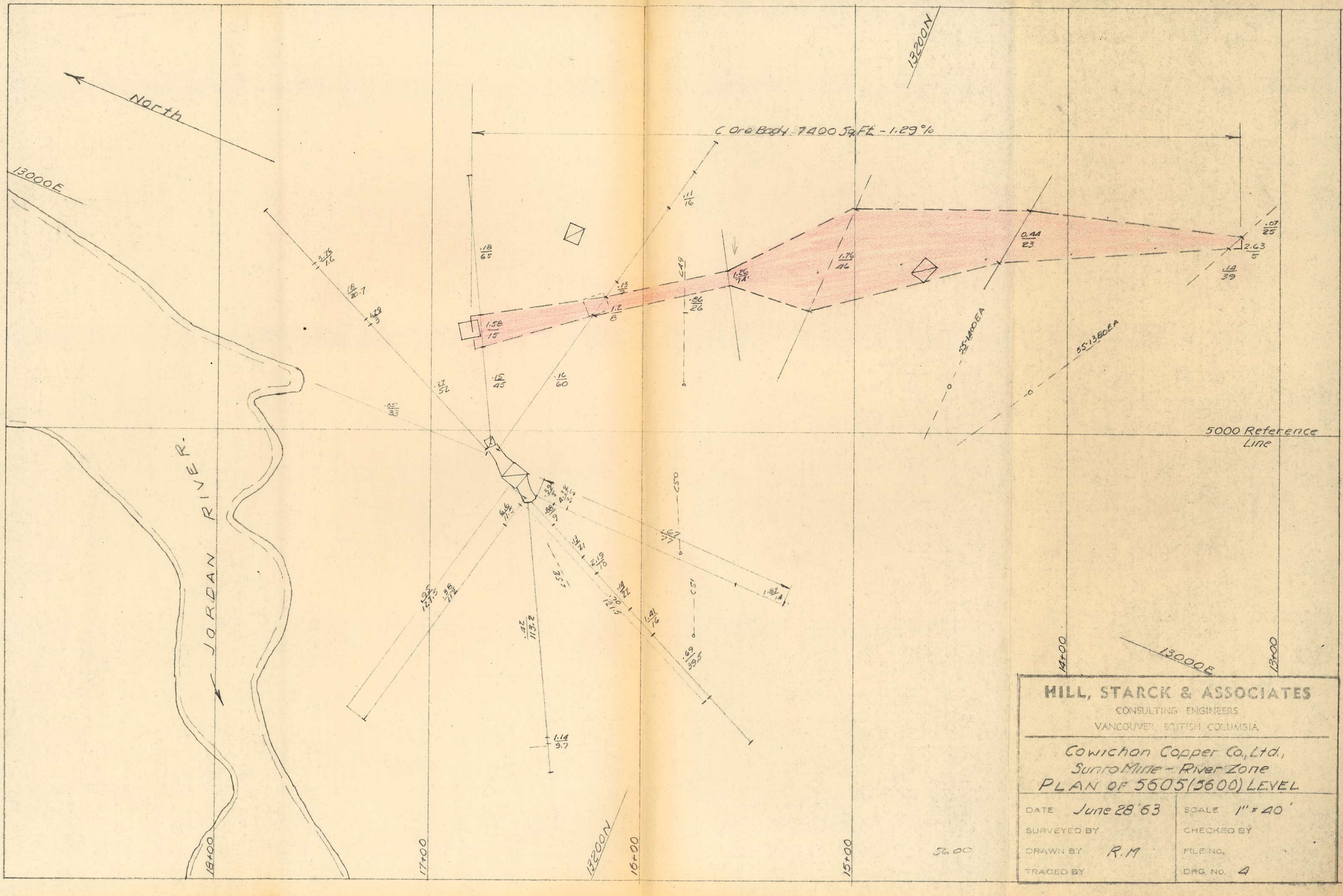
VANCOUVER, BRITISH COLUMBIA

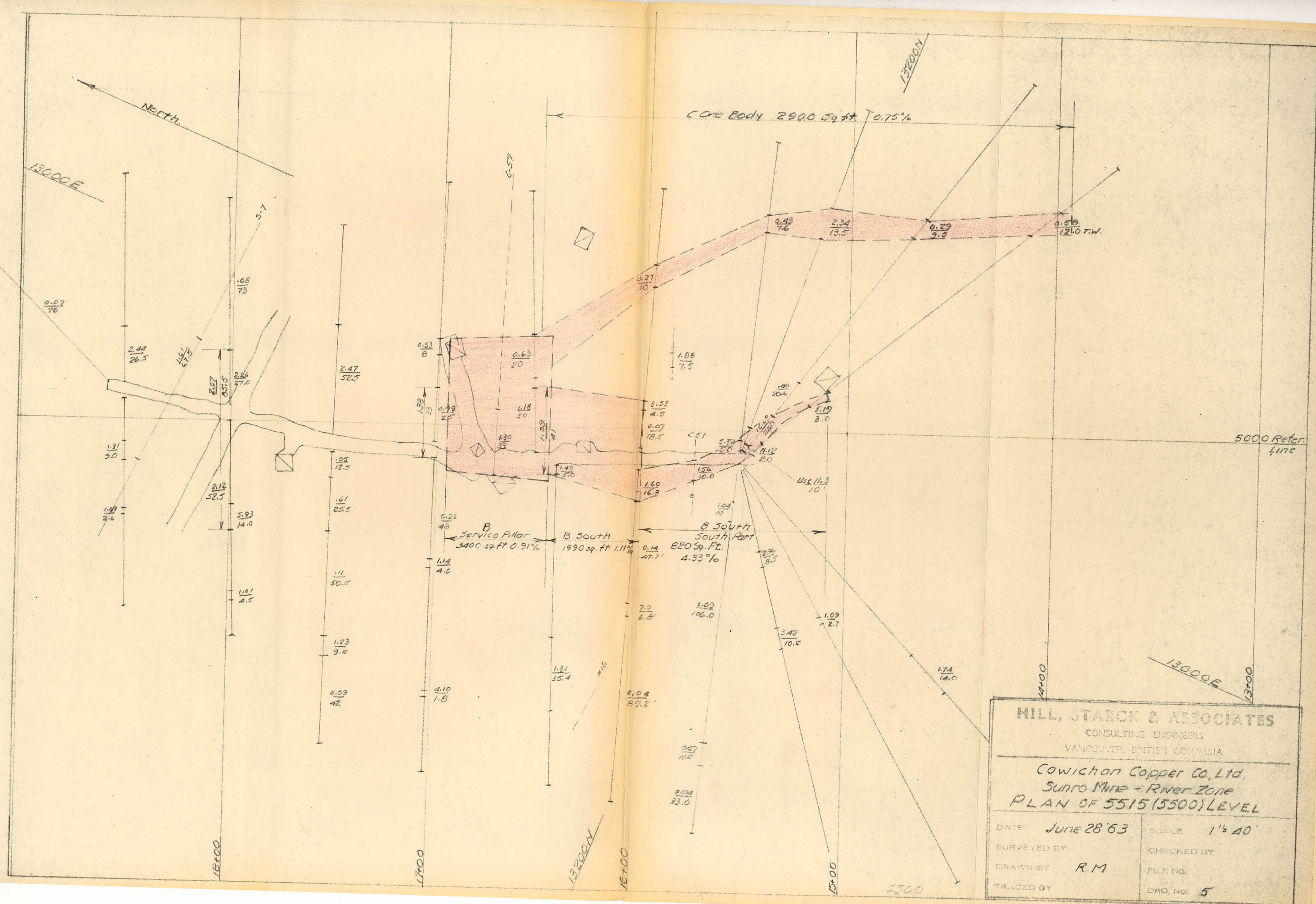
Cowichan Copper Co., Ltd.
Sunro Mine - River Zone
PLAN OF 5770 (5800) LEVEL

DATE June 28 '63	SCALE 1" = 40'
SURVEYED BY	CHECKED BY
DRAWN BY R.M	FILE NO.
TRACED BY	DRG. NO. 2

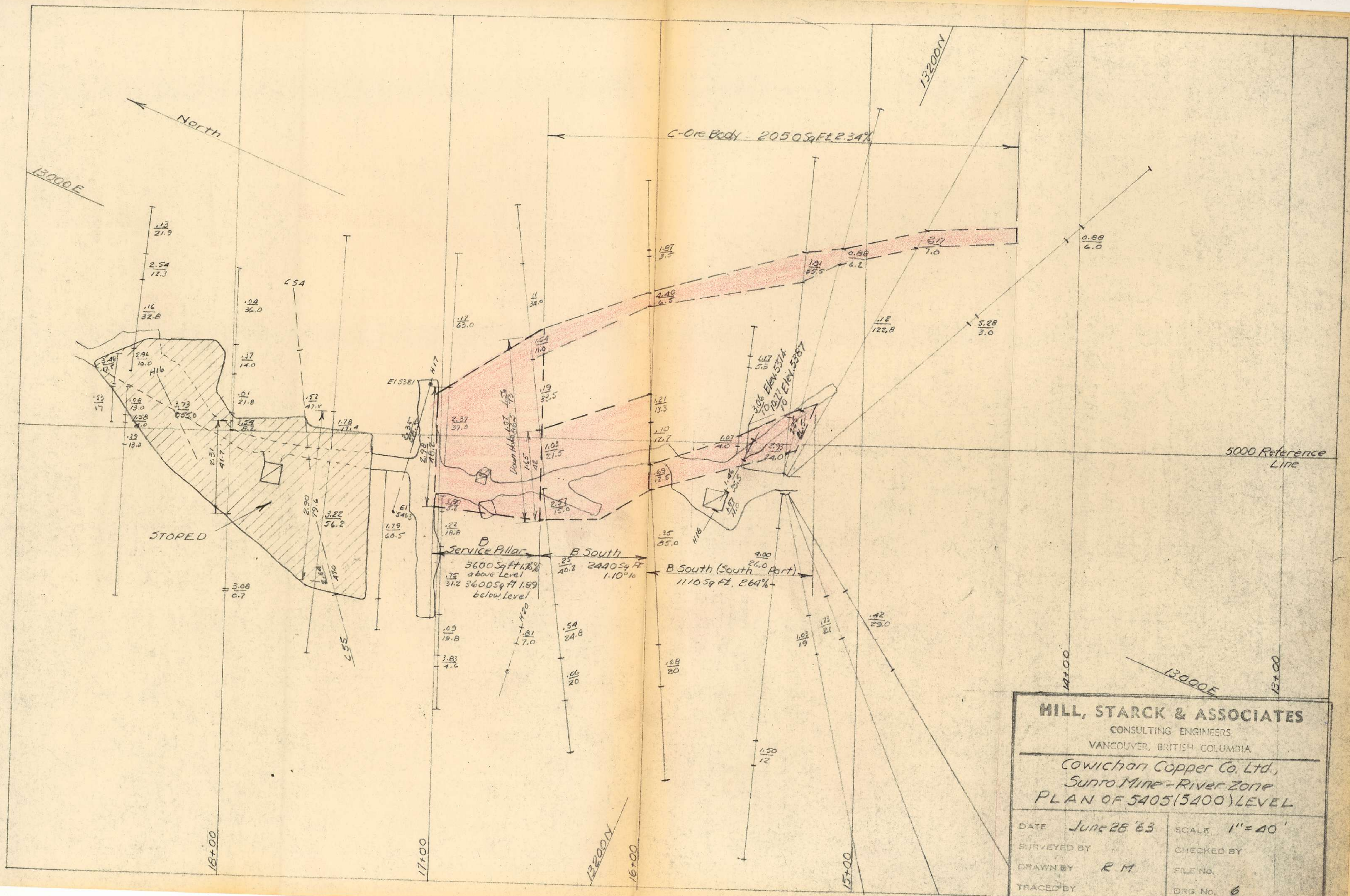
5800

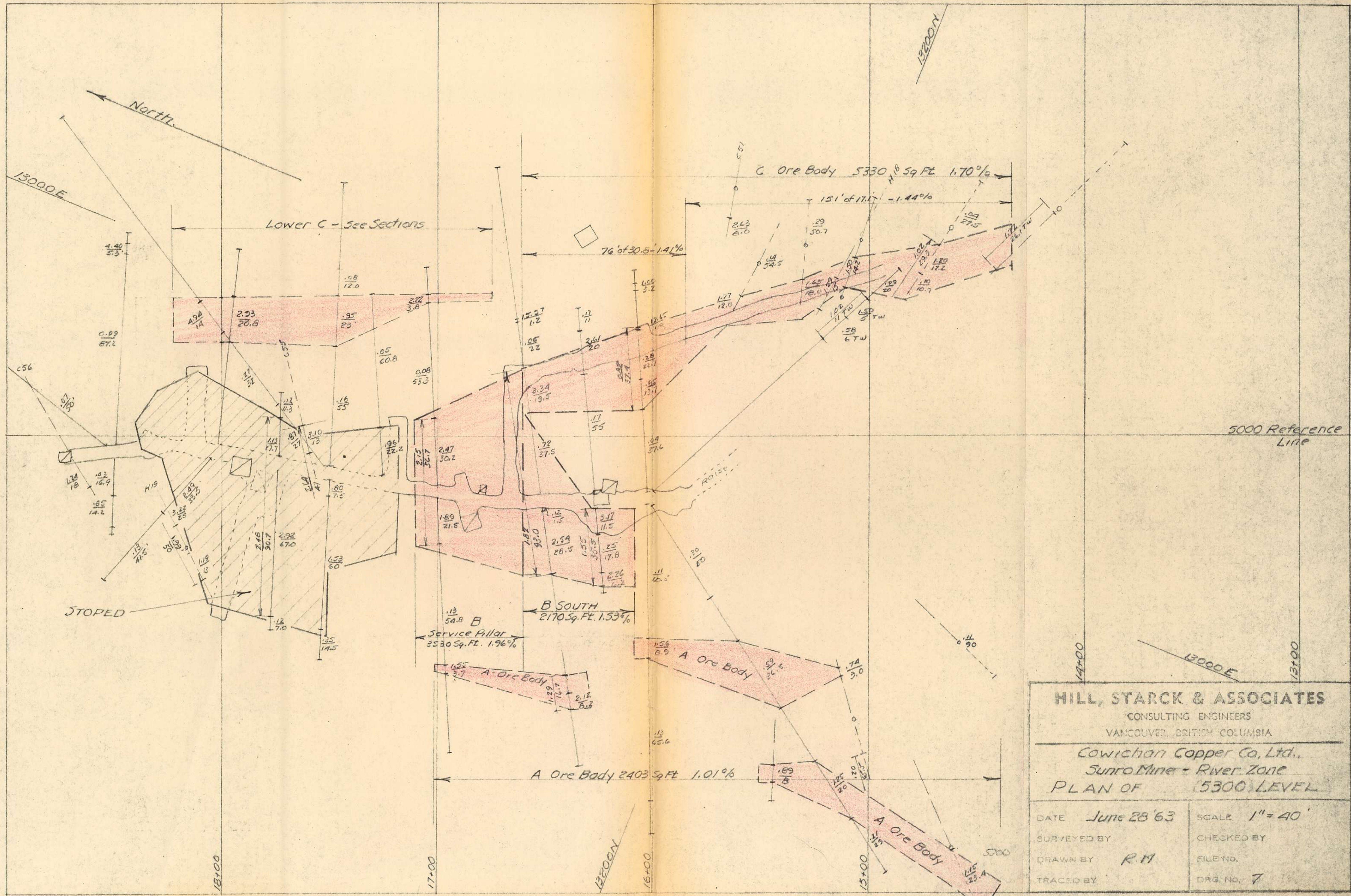

"C" Ore Body 10900 Sq.Ft.-2.08%
13200N
North
13000E
Channel Samples
5000 Reference Line
17+00
16+00
13200N
15+00
14+00
13+00
13000E
12+00
HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co. Ltd.
Sunro Mine - River Zone
PLAN OF 5670 (5700) LEVEL
DATE June 28 '63
SCALE 1" = 40'
SURVEYED BY
CHECKED BY
DRAWN BY R.M.
FILE NO.
TRACED BY
DRG. NO. 3
5670

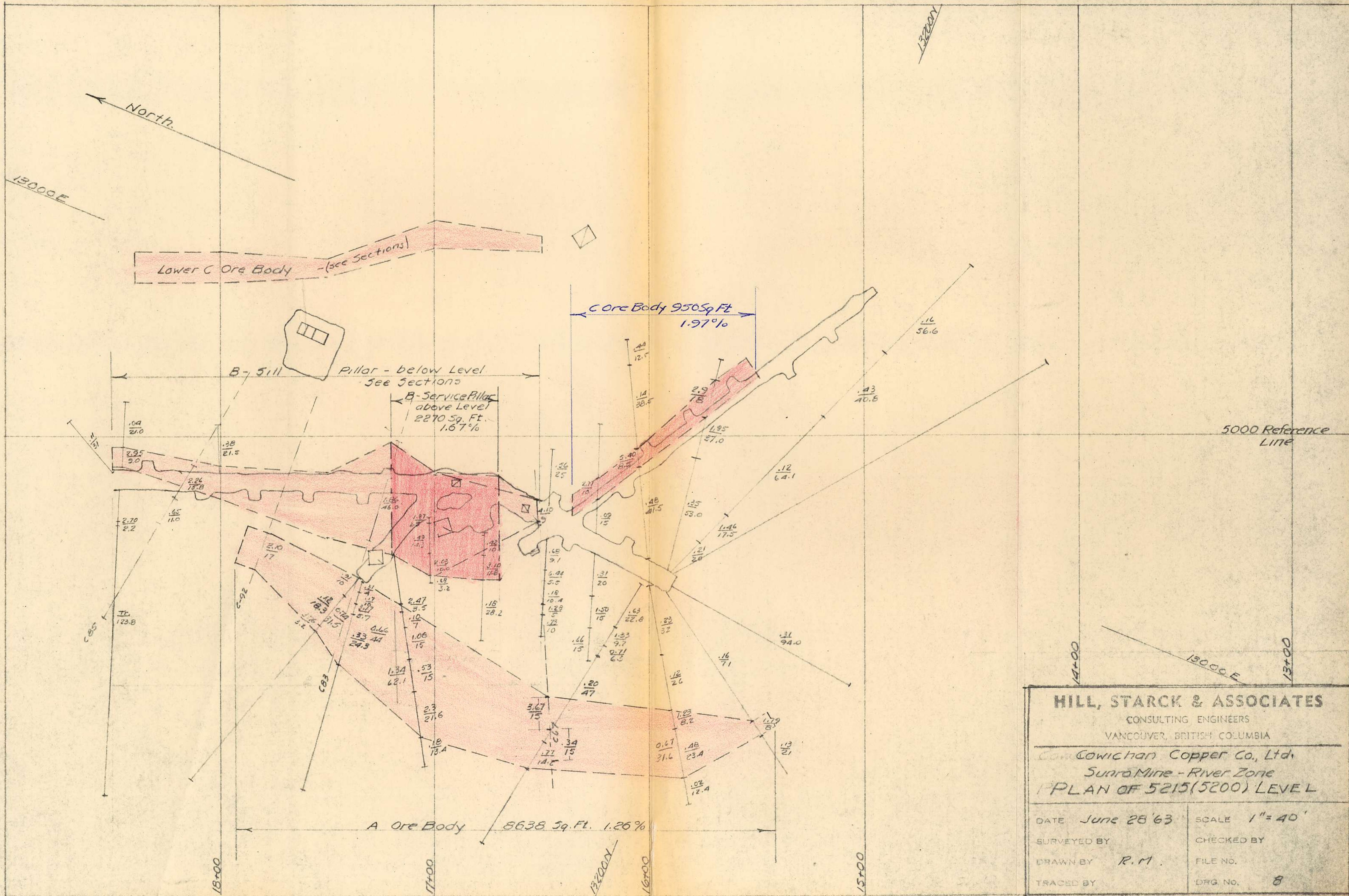

North
13000E
JORDAN RIVER
C Ore Body 7400 Sq Ft - 1.29%
13200N
5000 Reference Line
55-1400EA
55-1380EA
18+00
17+00
13200N
16+00
15+00
5600
14+00
13+00
13000E
HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co., Ltd.
Sunro Mine - River Zone
PLAN OF 5605(5600) LEVEL
DATE June 28 '63
SCALE 1" = 40'
SURVEYED BY
CHECKED BY
DRAWN BY R.M
FILE NO.
TRACED BY
DRG. NO. 4



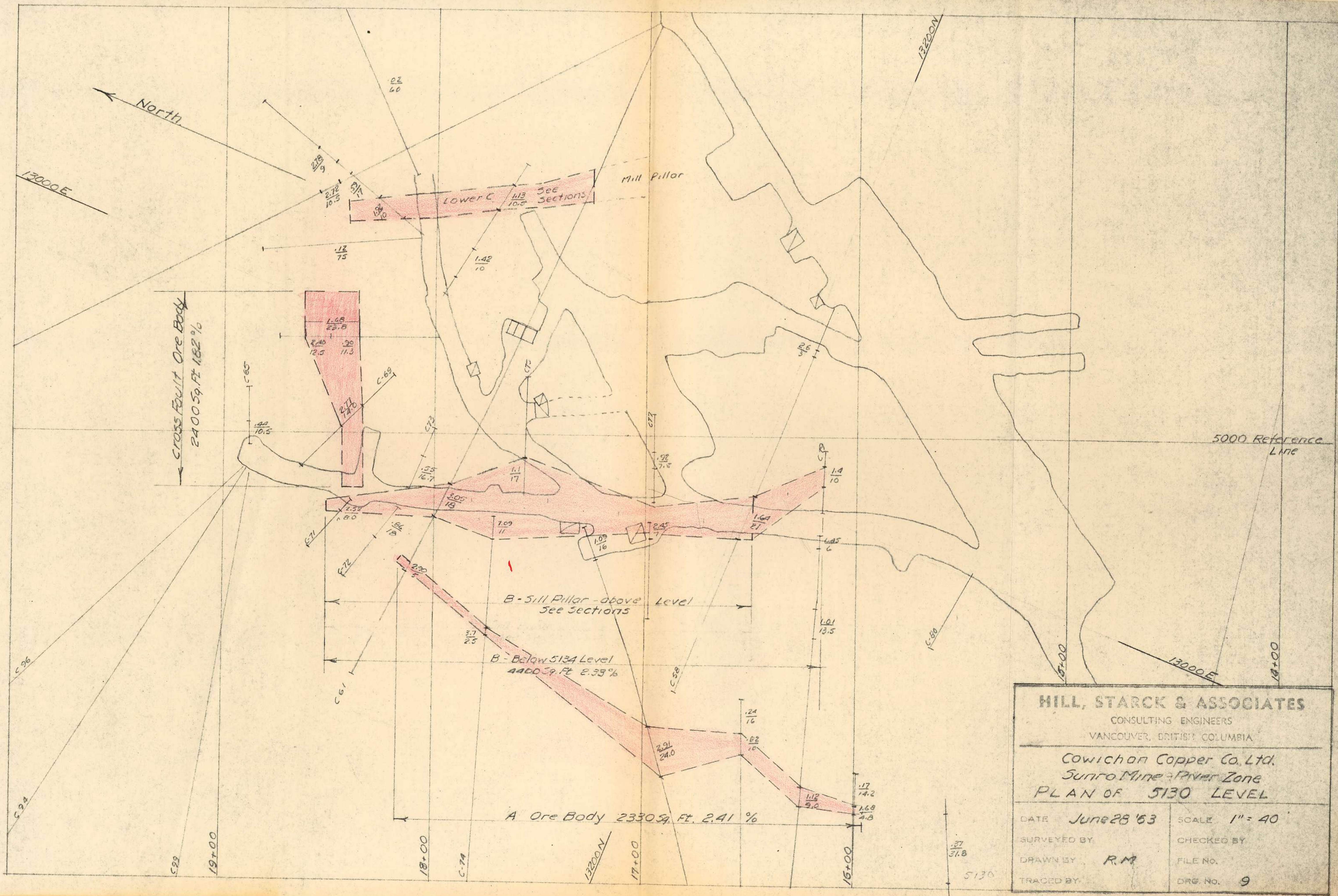
North
13000E
13200N
C Ore Body 2900 Sq.ft. 0.75%
B Service Pillar 3400 sq.ft 0.91%
B South 1990 sq.ft 1.11%
B South South Part 820 Sq.Ft. 4.33%
5000 Refer. Line
18+00
17+00
16+00
15+00
14+00
13+00
5500
HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co. Ltd,
Sunro Mine - River Zone
PLAN OF 5515 (5500) LEVEL
DATE June 28 '63
SCALE 1" = 40'
SURVEYED BY
CHECKED BY
DRAWN BY R.M
FILE NO.
TRACED BY
DRG. NO. 5


North
13000E
13200N
C-Ore Body 2050 Sq Ft 2.34%
5000 Reference Line
STOPED
H16
H17
H18
H20
C54
C55
E1 5381
E1 5463
Down Hole
3.06 Elev. 5314
1.02 Elev. 5387
B Service Pillar
3600 Sq Ft 1.76% above Level
3600 Sq Ft 1.89 below Level
B South
2440 Sq Ft 1.10%
B South (South Part)
1110 Sq Ft. 2.64%
18+00
17+00
16+00
15+00
14+00
13+00
HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co. Ltd.,
Sunro Mine - River Zone
PLAN OF 5405 (5400) LEVEL
DATE June 28 '63
SCALE 1" = 40'
SURVEYED BY
CHECKED BY
DRAWN BY R M
FILE NO.
TRACED BY
DRG. NO. 6

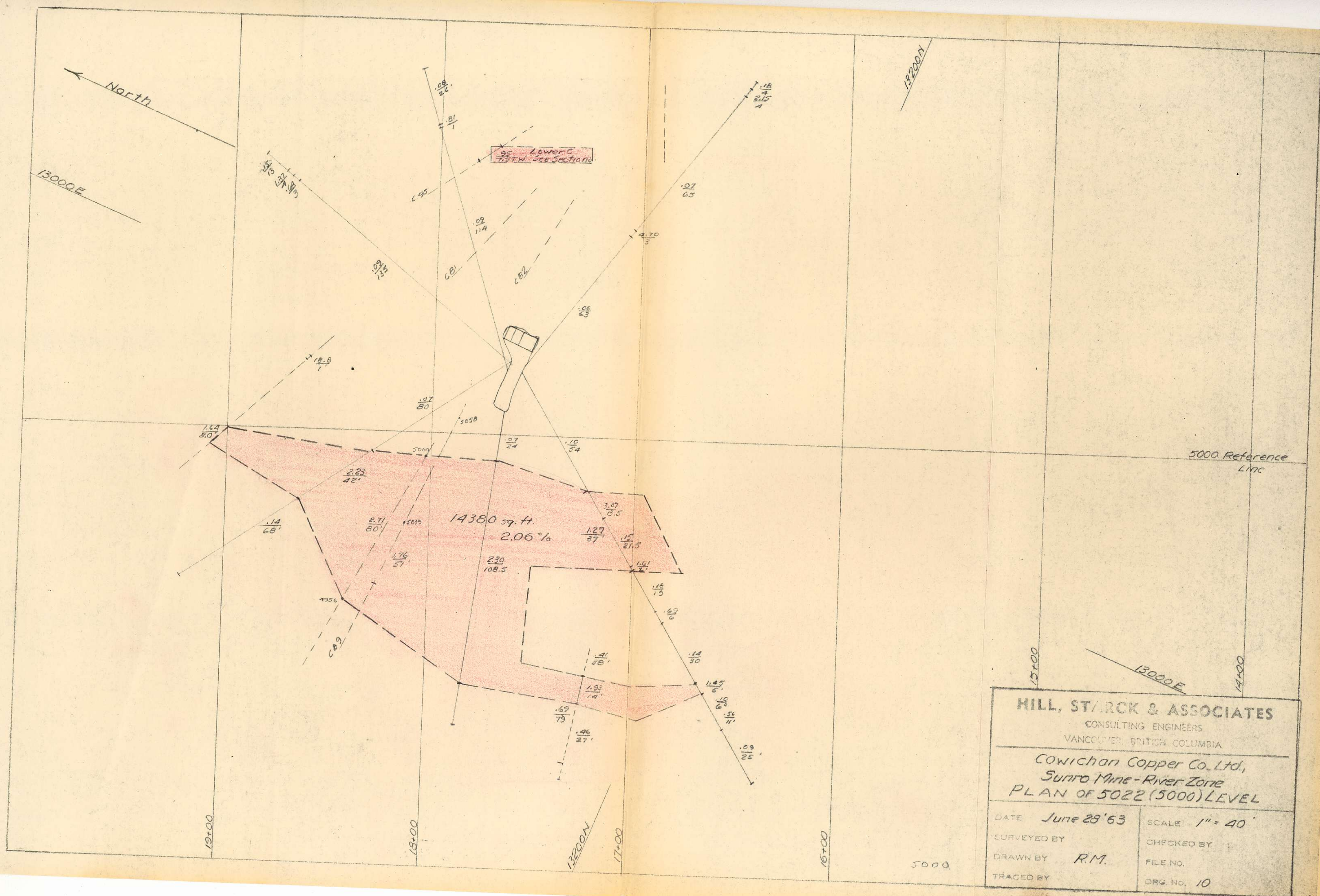

HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co, Ltd.
Sunro Mine - River Zone
PLAN OF 5300 LEVEL
DATE June 28 '63
SCALE 1" = 40'
SURVEYED BY
CHECKED BY
DRAWN BY R.M.
FILE NO.
TRACED BY
DRG. NO. 7
North
13000E
13200N
5000 Reference Line
C Ore Body 5330 Sq Ft 1.70%
151' of 17.1' - 1.44%
76' of 30.8' - 1.41%
Lower C - See Sections
B SOUTH
2170 Sq. Ft. 1.53%
B
Service Pillar
3530 Sq. Ft. 1.96%
A Ore Body
A Ore Body 2403 Sq Ft 1.01%
STOPED
Raise
18+00
17+00
16+00
15+00
14+00
13+00

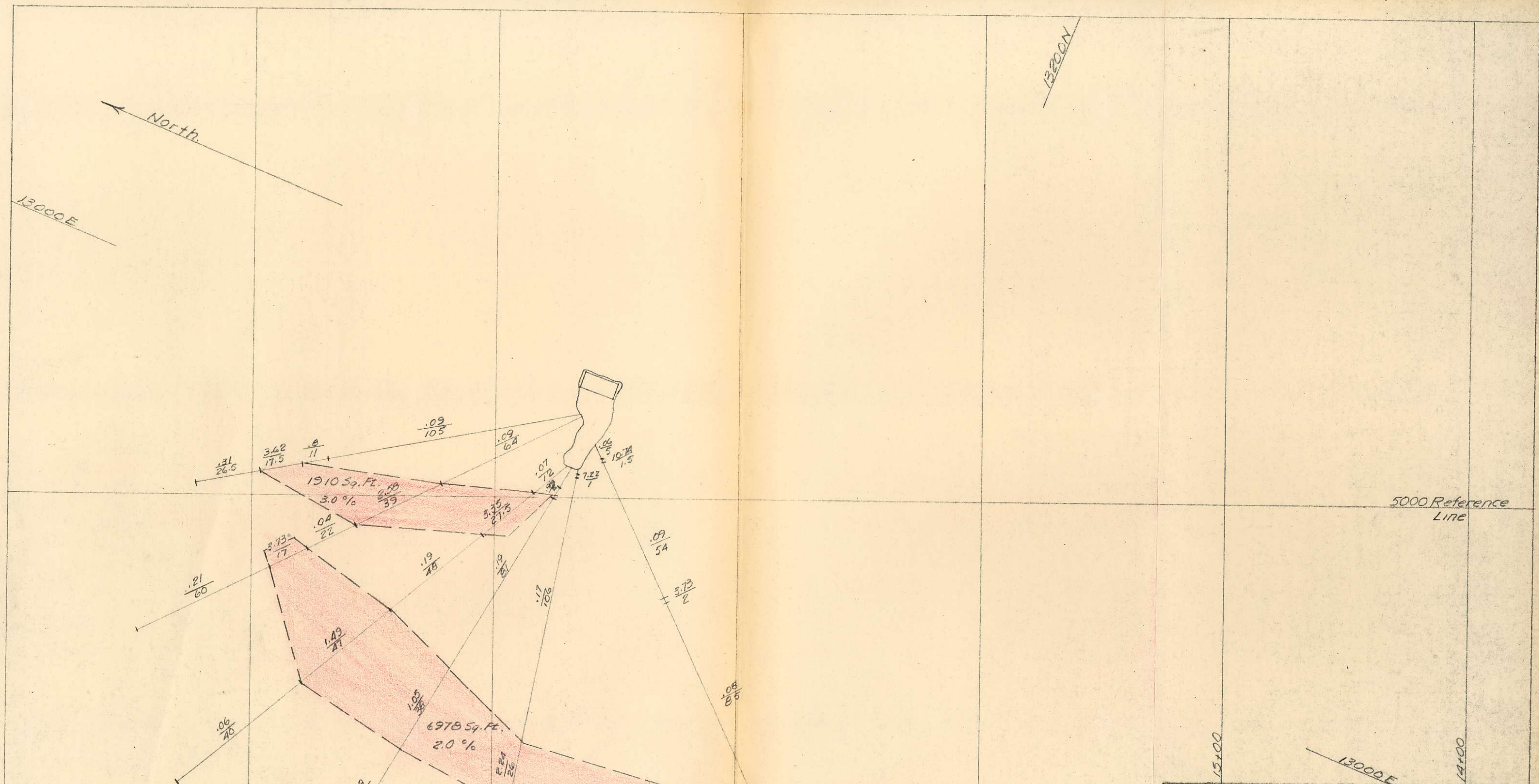


North.
13000E
Lower C Ore Body -(see Sections)
C Ore Body 950 Sq Ft
1.97%
B-Sill Pillar - below Level
See Sections
B-Service Pillar
above Level
2270 Sq. Ft.
1.67%
5000 Reference Line
13200N
A Ore Body 8638 Sq. Ft. 1.26%
18+00
17+00
16+00
15+00
14+00
13+00
13000 E
HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co., Ltd.
Sunro Mine - River Zone
PLAN OF 5215(5200) LEVEL
DATE June 28 '63
SCALE 1" = 40'
SURVEYED BY
CHECKED BY
DRAWN BY R.M.
FILE NO.
TRACED BY
DRG. NO. 8



North
13000E
13200N
Lower C
See Sections
Mill Pillar
Cross Fault Ore Body
2400 Sq. Ft. 1.82 %
5000 Reference Line
B - Sill Pillar - above Level
See Sections
B - Below 5134 Level
4400 Sq. Ft. 2.33 %
A Ore Body 2330 Sq. Ft. 2.41 %
19+00
18+00
17+00
16+00
15+00
14+00
C-65
C-69
C-71
C-72
C-73
C-74
C-77
C-58
C-61
C-80
C-93
C-94
C-96
HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co. Ltd.
Sunro Mine - River Zone
PLAN OF 5130 LEVEL
DATE June 28 '63
SCALE 1" = 40'
SURVEYED BY
CHECKED BY
DRAWN BY R.M.
FILE No.
TRACED BY
DRG. No. 9
5130


North
13000E
13200N
Lower C
75TW See Sections
C95
C81
C82
C89
14380 sq. ft.
2.06 %
5000 Reference Line
19+00
18+00
17+00
16+00
15+00
14+00
5000
HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co. Ltd.,
Sunro Mine - River Zone
PLAN OF 5022 (5000) LEVEL
DATE June 28 '63
SCALE 1" = 40'
SURVEYED BY
CHECKED BY
DRAWN BY R.M.
FILE NO.
TRACED BY
DRG. NO. 10



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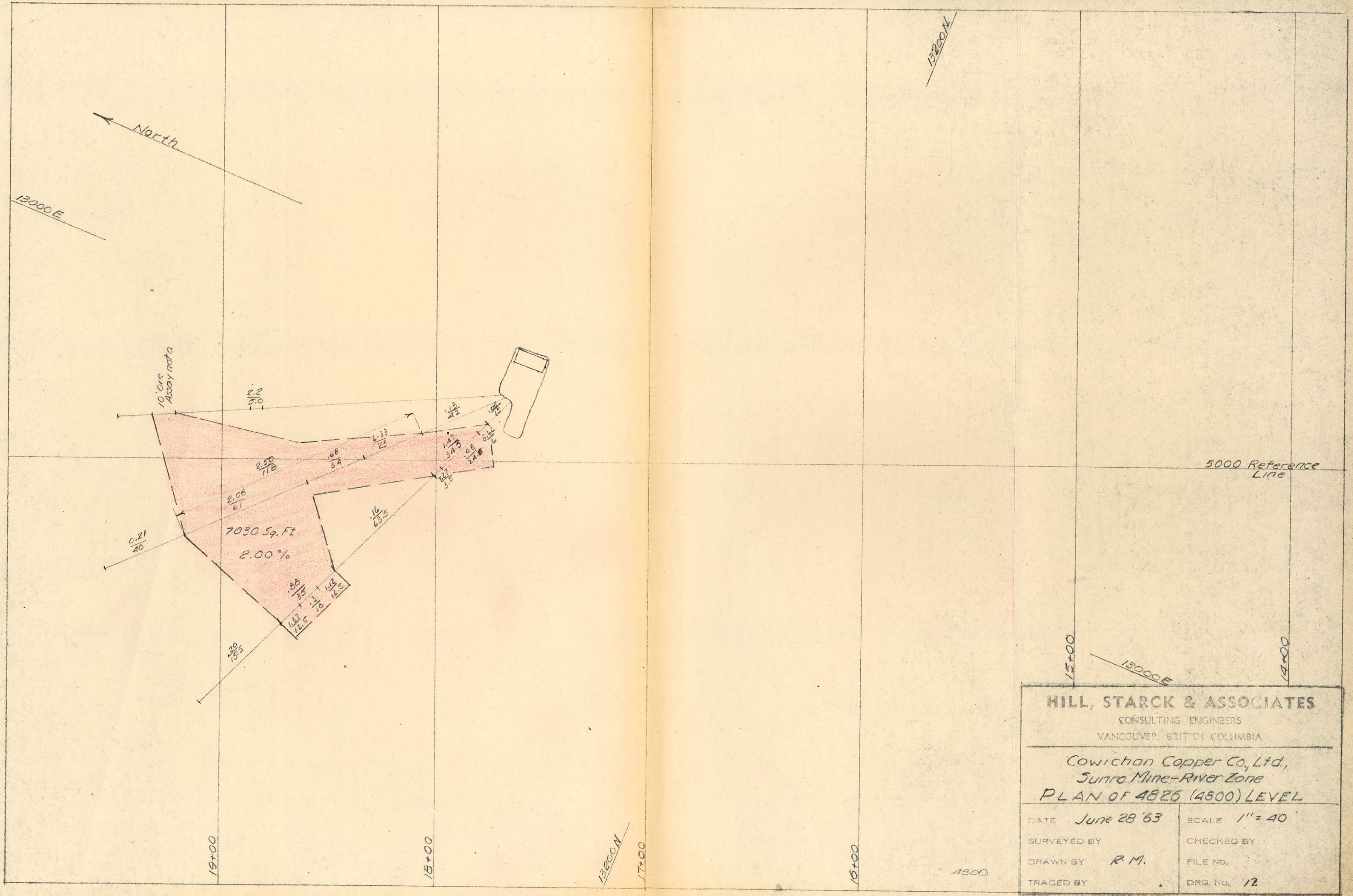
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.,
Sunro Mine - River Zone
PLAN OF 4904 (4900) LEVEL

DATE	June 28 '63	SCALE	1" = 40'
SURVEYED BY		CHECKED BY	
DRAWN BY	R. M.	FILE NO.	
TRACED BY		DRG. NO.	11



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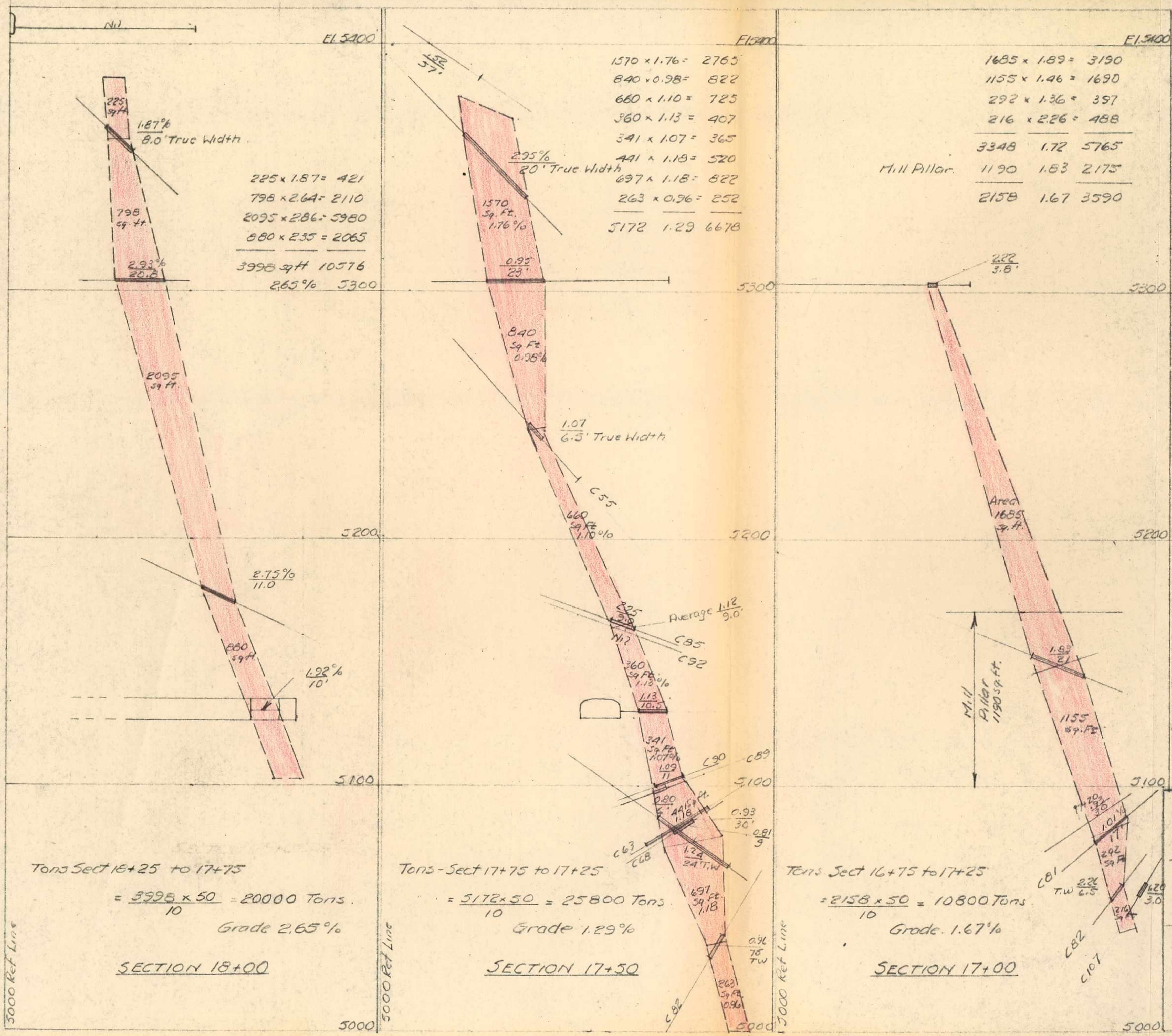
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.,
Sunro Mine-River Zone
PLAN OF 4826 (4800) LEVEL

DATE June 28 '63	SCALE 1" = 40'
SURVEYED BY	CHECKED BY
DRAWN BY R.M.	FILE NO.
TRACED BY	DRG. NO. 12



SUMMARY - Lower C.

Sect 17+00	10800 Tons	@ 1.67%	18040
17+50	25800 Tons	@ 1.29%	33300
18+00	20000 Tons	@ 2.65%	53000
Total	56600 Tons	@ 1.85%	104340

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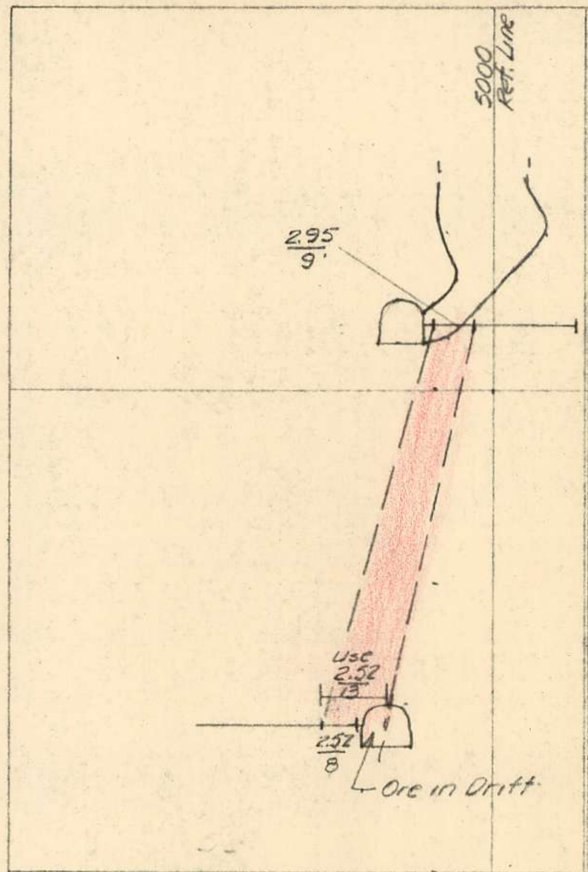
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co, Ltd.
Sunro Mine - River Zone
CROSS SECTIONS - LOWER "C" ORE BODY

DATE	June 28 '63	SCALE	1" = 40'
SURVEYED BY		CHECKED BY	
DRAWN BY	RM	FILE NO.	
TRACED BY		DRG. NO.	13.

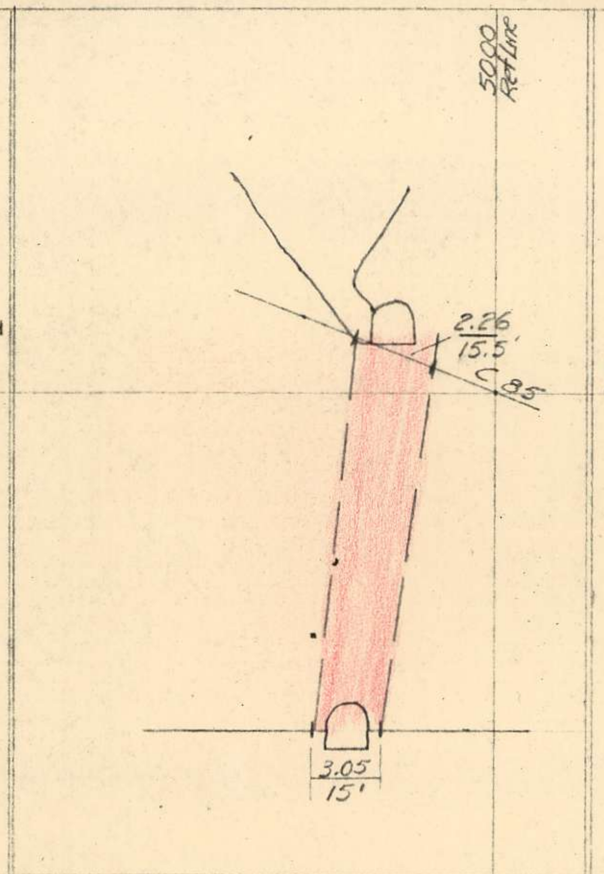


9 × 2.95 = 26.55
8 × 2.52 = 20.16
17 2.75 46.71

Area $\frac{17}{2} \times 84 = 710$ Sq.Ft

Grade 2.75%

SECTION 18+50

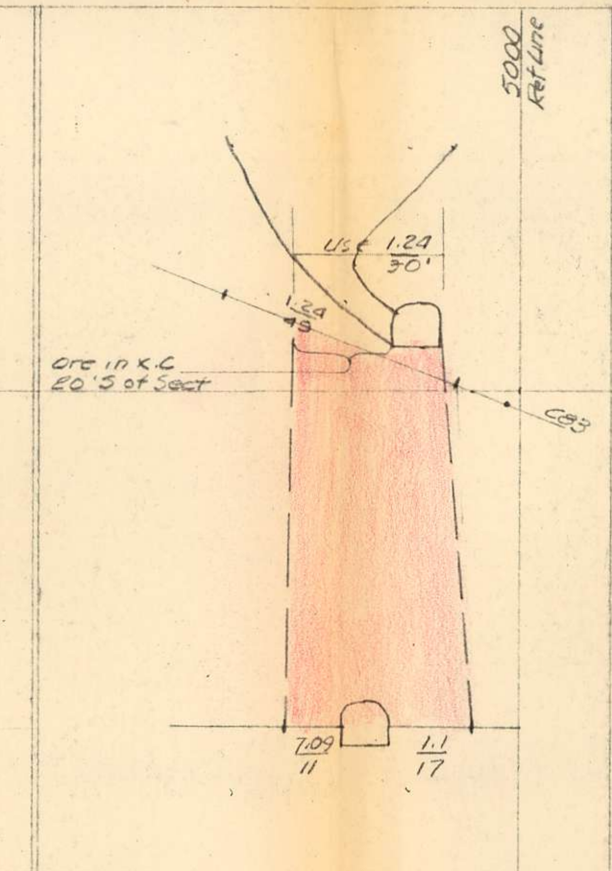


15.5 × 2.26 = 35.10
15 × 3.05 = 45.8
30.5 2.66 80.90

Area $\frac{30.5}{2} \times 84 = 1280$ Sq.Ft.

Grade 2.66%

SECTION 18+00

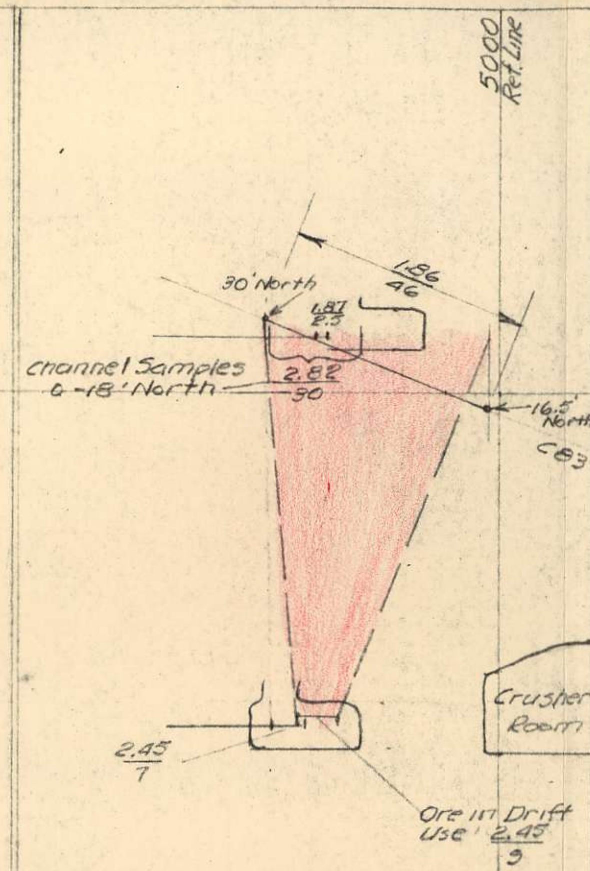


11 × 7.09 = 77.99
17 × 1.1 = 18.70
28 3.45 96.69
30 1.24 37.20
58 2.30 133.89

Area $\frac{30+39}{2} \times 84 = 2900$ Sq.Ft

Grade 2.30%

SECTION 17+50

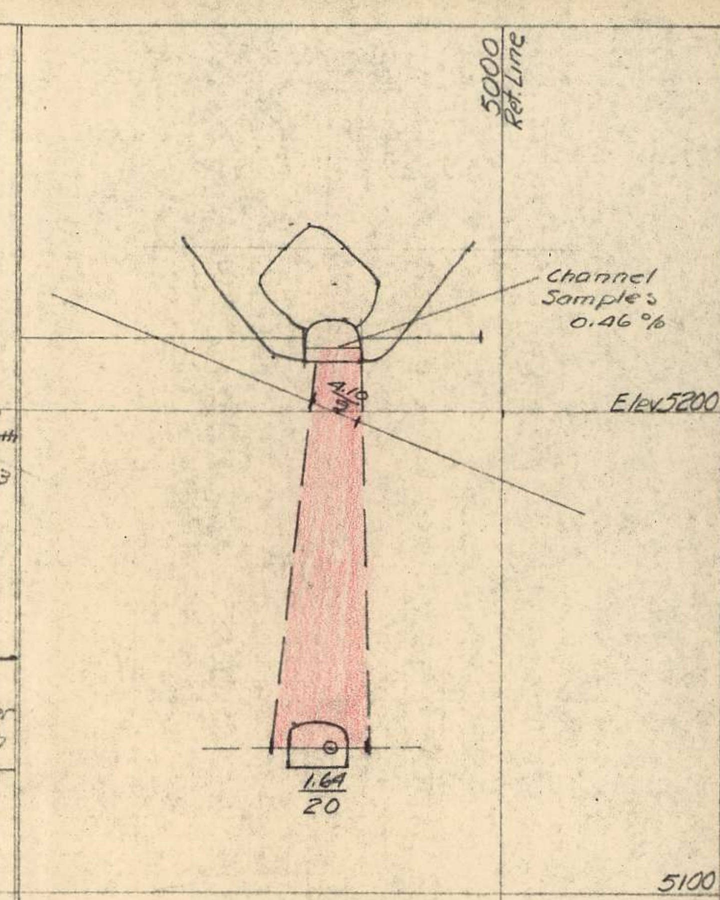


46 × 1.86 = 85.60
9 × 2.45 = 22.05
55 1.96 107.65

Area $\frac{55}{2} \times 84 =$ 2310 Sq.Ft.

Grade 1.96%

SECTION 17+00



20 × 1.64 = 32.80
9 × 2.28 = 20.52
29 1.84 53.32

Area $\frac{29}{2} \times 84 =$ 1220 Sq.Ft.

Grade 1.84%

SECTION 16+50

Sect 16+50	1220(1.84) = 2245	$\frac{6775}{3530}$ = 1.92%	$\frac{3530}{2} \times \frac{50}{10}$ = 8800 Tons	at 1.92%	16,900
Sect 17+00	2310(1.96) = 4530	$\frac{11200}{5210}$ = 2.15%	$\frac{5210}{2} \times \frac{50}{10}$ = 13000 "	at 2.15%	28000
Sect 17+50	2900(2.30) = 6670	$\frac{10070}{4180}$ = 2.41%	$\frac{4180}{2} \times \frac{50}{10}$ = 10500 "	at 2.41%	25300
Sect 18+00	1280(2.66) = 3400	$\frac{5350}{1990}$ = 2.69%	$\frac{1990}{2} \times \frac{50}{10}$ = 5000 "	at 2.69%	13400
Sect 18+50	710 (2.75) = 1950				
		TOTAL	37300 Tons	at 2.24%	83600

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CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co. Ltd.
Sunro Mine – River Zone
CROSS SECTIONS – SILL PILLAR – B ORE BODY

DATE June 28 '63	SCALE 1" = 40'
SURVEYED BY	CHECKED BY
DRAWN BY R.M.	FILE No.
TRACED BY	DRG. No. 14

River Zone - C Ore Body

5770 (5800) Level — 170'

Drill Hole	4.7' - 2.05%		9.65
" "	2.5' - 0.79		1.98
" "	12.0 - 2.03		24.36
Channel	14.0 - 2.43		34.02
"	15.0 - 1.80		27.00
Drill Hole	22.0 - 0.9		19.80
Channel	15 - 0.9		13.50
Drill Hole	17 - 0.85		3.14
" "	15.5 - 1.78		27.60
" "	13.5 - 1.25		16.90
" "	12 - 1.83		21.96
	143.2	1.40	199.91

5770 Area 5480 Sq. Ft

Holes above 5770 to 5940

7.0	1.60	11.20
10.0	2.51	25.10
24.0	1.40	33.60
20.0	0.64	12.80
12.0	1.09	13.08
16.0	1.48	23.60
× 4.0	4.03	16.12
93.0	1.46	135.50

5940 Area. 2200 Sq. Ft

$\frac{5480+2200}{2} \times \frac{170}{10}$ = 65000 Tons Grade 1.42%

5670 (5700) Level. Channel Sample

50 × 11.5 × 1.07	615
" 25 × 25 × 0.86	537
" 28 × 39 × 1.07	1165
" 24 × 43 × 1.76	1815
" 25 × 32 × 0.97	776
" 25 × 25 × 3.12	1985
" 25 × 35 × 1.55	1358
" 34 × 25 × 2.65	2255
" 34 × 30.5 × 2.33	2350
" 25 × 33.5 × 1.83	1531
" 30 × 28 × 2.91	2445
" 60 × 20 × 3.91	4690
2.08%	

5670 Area. 10900 Sq. Ft

$\frac{5480+10900}{2} \times \frac{100}{10}$ = 81900 Tons Grade 1.85%

5605 (5600) Level Drill Hole

37 × 15	555	× 1.58	877
" 60 × 8	480	× 1.20	576
" 56 × 7.4	414	× 1.26	522
" 65 × 46	2990	× 1.76	5262
" 96 × 23	2208	× 0.44	972
" 53 × 5	265	× 2.63	6.97
367	6912	1.29	8906

5600 Area 7400 Sq Ft

$\frac{10900+7400}{2} \times \frac{65}{10}$ = 59500 Tons Grade 1.76%

Holes between 5515 and 5605 Levels Drill Hole

97 × 10	970	× 0.8	775
" 68 × 12	816	× 1.32	1077
" 134 × 26	3480	× 0.86	2995
299	5266	0.92	4847

$\frac{7400+2900}{2} \times \frac{90}{10}$ = 46300 Tons Grade. 0.99%

5515 (5500) Level Drill Hole

20 × 20	400	× .63	252
55 × 10	550	× .27	149
42 × 7.6	319	× .43	137
38 × 13.5	512	× 2.34	1190
47 × 9.5	446	× 0.29	129
33 × 12.5	412	× 0.44	182
24 × 12	288	× 0.58	167
259	2927	0.75	2206

5500 Area 2900 Sq Ft

$\frac{2900+2050}{2} \times \frac{110}{10}$ = 27200 Tons Grade 1.40%

Holes between 5405 and 5515 Levels Drill Hole

74 × 21	1554	× 1.3	2020
67 × 11	737	× 1.8	1326
91 × 11.5	1048	× .33	346
232 144	3339	1.11	3692

5405 (5400) Level Drill Hole

22 × 11	242	1.54	373
" 64 × 6.5	416	4.4	1830
" 44 × 13.5	594	1.91	1135
" 29 × 6.2	180	0.88	159
" 56 × 7.5	420	2.17	921
" 10 × 6.5	65	0.88	57
225 8.5	1917	2.34	4475

5400 Area. 2050 Sq Ft.

$\frac{2050+5330}{2} \times \frac{105}{10}$ = 38700 Tons Grade 2.13%

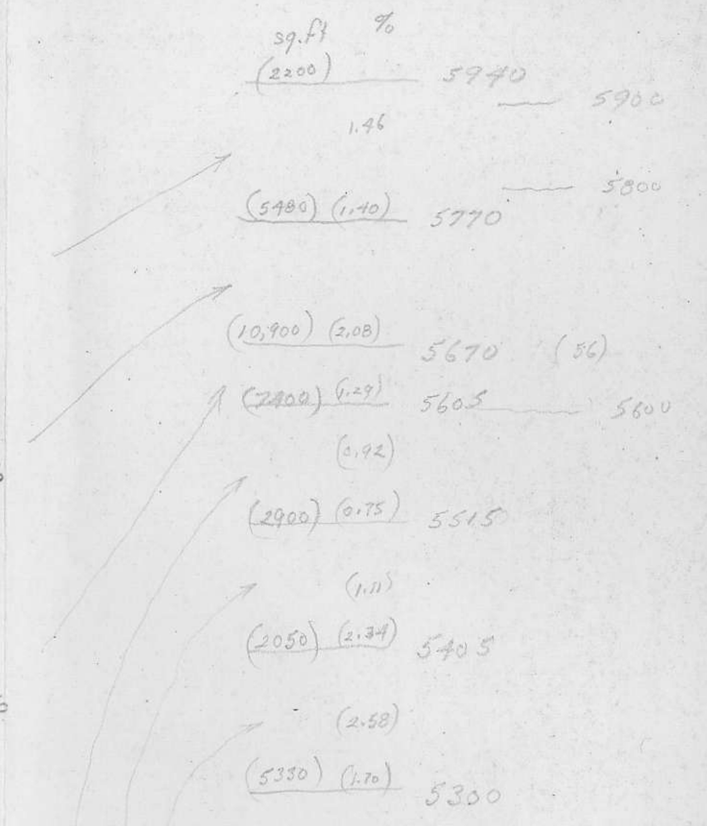
Holes between 5300 and 5405 Levels Drill Hole

13	×	6.86	89.1
29	×	0.65	18.9
42		2.58	108.0

5300 Level Drill Hole

19.5	3.34	65.1
" 20	2.61	52.2
" 37.4	.82	30.7
" 12.0	1.77	21.3
" 18.0	1.65	29.7
" 14.2	1.50	21.3
" 22	1.02	22.4
" 26.1	1.72	44.8
169.2	1.70	287.5

Area. 5330 Sq. Ft



Continued on Map 16

HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.
Sunro Mine - River Zone
Ore Reserves - Upper C Ore Body

DATE June 28 '63	SCALE —
SURVEYED BY	CHECKED BY
DRAWN BY R.M	FILE NO.
TRACED BY	DRG. NO. 15

River Zone - C Ore Body (Cont'd)

5215 Level.

10'	2.77%	27.70
8.5	5.40	45.90
18	2.90	52.20
36.5	3.44%	125.80

Area 950 Sq. Ft

$\frac{5330+950}{2} \times \frac{65}{10}$ = 20,400 Tons Grade 1.97%

TOTAL 339,000 Tons Grade 1.64%

River Zone - B Ore Body (Below 5130)

5134 Level.

208 Sq Ft	2.52	524
616	3.05	1880
1150	3.45	3970
952	1.09	1040
560	2.45	1370
720	1.64	1180
187	1.4	262
4393	2.33	10226

Area 4400 Sq. Ft. 2.33%

$\frac{18780}{2} \times \frac{111}{10}$ = 104,200 Tons. Grade 2.13%

5022 Level.

582	1.64	955
1729	2.23	3860
3430	2.23	7640
5395	2.30	12420
2075	1.27	2650
870	1.93	1680
300	1.45	435
14380	2.06	29640

Area. 14380 Sq Ft 2.06%

$\frac{23269}{2} \times \frac{118}{10}$ = 137,200 Tons Grade 2.12%

290,864

4904 Level

330	3.62	1193
973	2.58	2510
607	3.35	2030
627	3.73	2340
1701	1.49	2535
1625	1.05	1705
2175	2.24	4870
288	5.45	1570
562	1.68	945
8888	2.24	19698

Area. 8888 Sq. Ft 2.24%

2.12

$\frac{15918}{2} \times \frac{78}{10}$ = 62,100 Tons. Grade 2.12%

199,300

131,652

422,516

4826 Level.

118 ft	2.50	295.0
34.3	1.43	49.2
35.0	0.88	31.0
187.3	2.00	375.2

Area 7030 Sq. Ft 2.00%

TOTAL 303,500 Tons Grade 2.12%

Indicated below 4826 Level.

$\frac{7030 \times 50}{10}$ = 35,100 Tons Grade 2.00%

HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER BRITISH COLUMBIA

Cowichan Copper Co., Ltd.
Sunro Mine - River Zone - Ore Reserves
Upper C Ore Body & B Ore Body below 5130

DATE	June 28 '63	SCALE	—
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DRAWN BY	R M.	FILE No.	
TRACED BY		DWG. No.	16

River Zone - B Ore Body

5560 Level — Area 0 Sq. Ft.

Service Pillar

5519 Level

33' × 1.38 = 45.5
15' × 0 = –
8 × 0.53 = 4.24
56' – 0.89 49.74

C 57
33' × 1.30 = 42.9
25' × 0.53 = 13.3
58 – 0.97 = 56.2

25' × 0.53 = 13.3
41' × 1.09 = 44.7
66' × 0.88 58.0
180 – 0.91 163.94

Area 3400 Sq. Ft 0.91%

$\frac{3400}{2} \times \frac{41}{10}$ = 7000 Tons Grade 0.91%

$\frac{7000}{2} \times \frac{116}{10}$ = 40600 Tons Grade 1.35%

5403 Level

48.2 × 2.98 = 143.5

42 × 1.65 = 69.3
33.5 × 0.19 = 6.4
11 × 1.54 = 16.9
86.5 1.07 92.6

Down Hole E
41 × 0.95 = 38.9
30 × .10 = 3.0
13 × 6.86 = 86.5
Down Hole W
11 1.82 20.0
95' 1.56 148.2 73' Horiz

Above Level 134.7 1.76 236.1
Below Level 128.0 1.89 241.9

Area 3600 Sq Ft 1.76 above
1.89 below

$\frac{7130}{2} \times \frac{99}{10}$ = 35400 Tons Grade 1.98%

5304 Level

21.5	1.89	40.6
5	1.40	7.0
30.2	2.47	74.6
56.7	2.15	122.2
93	1.82	169.2
149.7	1.96	291.3

Area 3530 Sq. Ft 1.96%

$\frac{5800}{2} \times \frac{89}{10}$ = 25800 Tons Grade 1.84%

5215 Level

2.5	1.87	4.67
14.3	0.49	7.00
10	2.03	20.3
26.8	1.19	31.97
10	0.42	4.2
11.8	3.1	36.6
21.8	1.87	40.8
46	1.86	85.5
26.8	1.19	31.9
21.8	1.87	40.8
94.6	1.67	158.2

Area 2270 Sq. Ft. 1.67%

TOTAL — 108,800 Tons Grade 1.62%

B South

5515 Level

4.5	3.53	15.9
18.5	0.7	13.0
10	0	–
16.3	1.6	26.1
49.3	1.12	55.0
41.0	1.09	44.6
90.3	1.11	99.8

Area 1990 Sq Ft 1.11%

$\frac{4430}{2} \times \frac{110}{10}$ = 24,400 Tons Grade 1.10%

5405 Level

13.3	1.21	16.1
12.7	.10	1.3
12.5	.69	8.6
6.5	0	–
45.0	.58	26.0
42.0	1.65	69.3
87	1.10	95.4

Area. 2440 Sq. Ft 1.10%

$\frac{4610}{2} \times \frac{105}{10}$ = 24200 Tons Grade 1.30%

5300 Level

11.5	3.17	36.4
17.8	0.25	4.5
6.2	2.26	14.0
35.5	1.55	54.9
28.5	2.54	72.3
1.5	0.12	0.2
7.0	0.00	0.0
21.0	0.72	15.1
58.0	1.51	87.6
93.5	1.53	142.5

Area. 2170 Sq. Ft. 1.53%

TOTAL — 48600 Tons Grade 1.20%

HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.,
Sunro Mine - River Zone - Ore Reserves
B Ore Body - Service Pillar & B South

DATE	June 28 '63	SCALE	–
SURVEYED BY		CHECKED BY	
DRAWN BY	RM	FILE NO.	
TRACED BY		DRG. NO.	17

River Zone - A Ore Body

5300 Level

223	Sq.Ft	1.55%	345.7	
562	"	1.29%	725.0	
387	"	1.56%	603.7	Elev. 5352-5257 $\frac{95 \times 2403}{10}$ = 23100 Tons 1.01%
1039	"	0.59%	613.0	
192	"	0.74%	142.1	
2403	Sq.Ft	1.01%	2429.5	

5215 Level

686	Sq Ft	2.10%	1440.6	
656		0.74	485.4	
872		0.66	575.5	
2208		1.34	2958.7	Elev. 5257-5173 $\frac{84 \times 8638}{10}$ = 72600 Tons 1.26%
2427		1.61	3907.5	
1473		0.67	986.9	
316		1.79	565.6	
8638		1.26	10920.2	

5130 Level

120	Sq.Ft	2.90%	348	
380		3.70	1406	
1060		2.91	3085	Elev 5173-5130 $\frac{43 \times 2327}{10}$ = 10000 Tons 2.41%
423		.82	347	
254		1.12	284	
90		1.68	151	
2327		2.41	5621	

TOTAL — 105700 Tons 1.31%

River Zone - B South Ore Body - South Part

5515 Level

16.3 × 1.6 = 26.1
10 × 1.56 = 15.6
10 × 11.3 = 113.0
3 × 5.19 = 15.6
39.3 4.33 170.3

Area ~~820~~ Sq.Ft. Elev. 5540-5460 $\frac{820 \times 80}{10}$ = 6560 Tons 4.33%

5405 Level

12.5 × .69 = 8.6
4 (1.07)
10 (3.06) 8.0 × 5.92 = 47.4
10 (10.77)
24 (2.93) 25.2 × 2.57 = 64.8
265 (2.25)
45.7 2.64 120.8

Area 1110 Sq.Ft Elev 5460-5338 $\frac{1110 \times 122}{10}$ = 13540 Tons 2.64%

TOTAL. — 20100 Tons 3.20%

River Zone - Cross Fault Ore Body

5130 Level

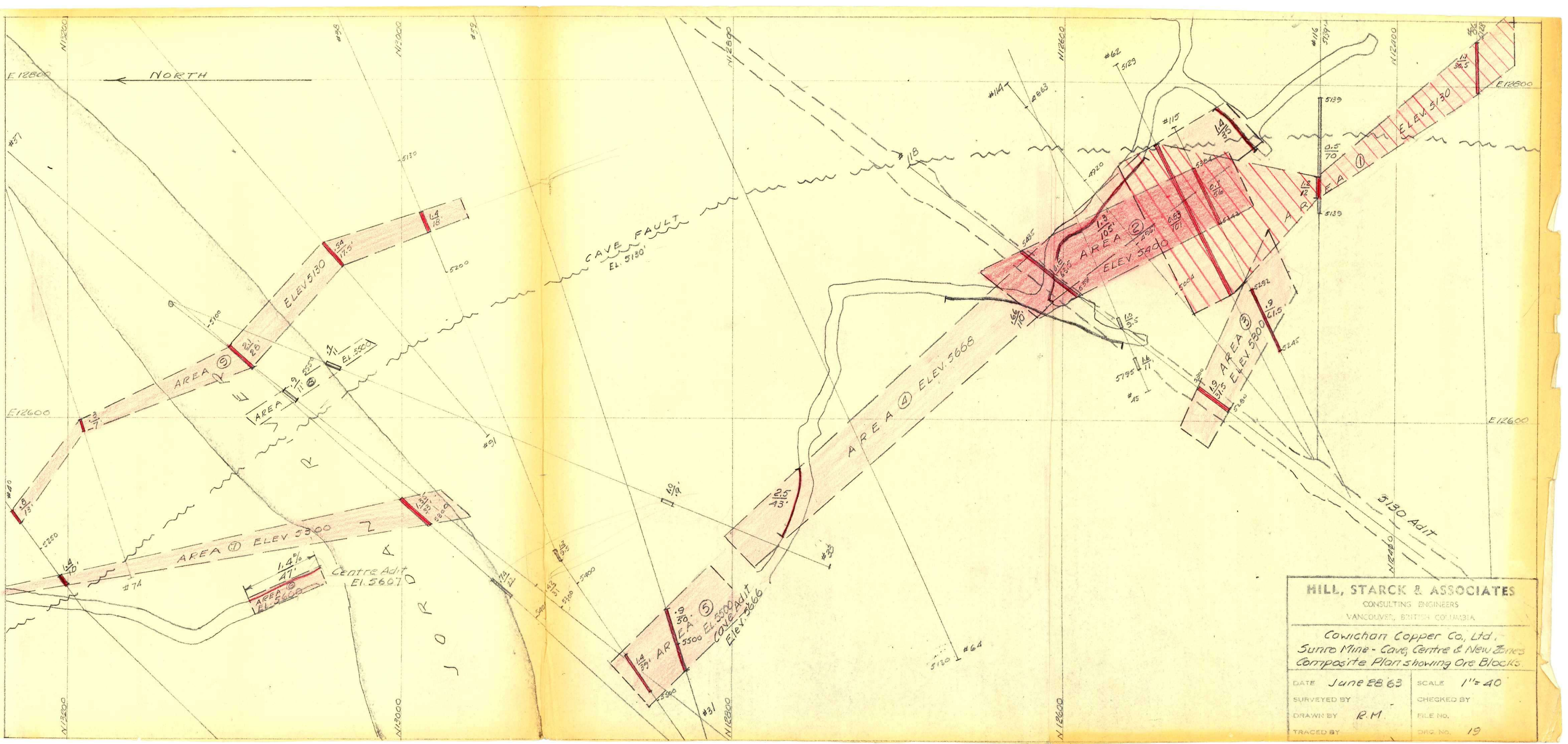
23.8 × 1.68 = 40.0
14 × 2.77 = 28.8
37.8 1.82 68.8

Area 2400 Sq Ft Elev 5180-5080 $\frac{2400 \times 100}{10}$ = 24000 Tons 1.82%

HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co. Ltd.
Sunro Mine - River Zone - Ore Reserves
A Ore Body, South Part of B South Ore Body and Cross Fault Ore Body

DATE June 28 '63	SCALE —
SURVEYED BY	CHECKED BY
DRAWN BY RM	FILE NO.
TRACED BY	DRG. NO. 18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CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA
Cowichan Copper Co., Ltd.
Sunro Mine - Cave, Centre & New Zones
Composite Plan showing Ore Blocks
DATE June 28 63
SCALE 1" = 40
SURVEYED BY
CHECKED BY
DRAWN BY R.M.
FILE No.
TRACED BY
DWG. No. 19
NORTH
E12800
E12600
N13200
N13000
N12800
N12600
N12400
CAVE FAULT
El. 5130'
AREA ① ELEV 5130
AREA ② ELEV 5400
AREA ③ ELEV 5300
AREA ④ ELEV 5668
AREA ⑤ ELEV 5130
AREA ⑤ El. 5500
AREA ⑦ ELEV 5300
AREA ⑥ El. 5600
AREA ⑥ El. 5500
Centre Adit El. 5607
Cave Adit Elev. 5666
5130 Adit
JORDAN RIVER

Cave Zone

Area ①	Elev. 5130	D.D. Hole 86	30.5'	1.1%	33.60	Area = 9500 Sq. Ft	$9500 \times \frac{100}{10}$ = 95,000 Tons	0.92%
		116	12.0	1.2	14.40			
		62	101.0	0.83	83.80			
		Average	143.5	0.92%	131.80			
Area ②	Elev. 5400	D.D. Hole 115	56'	0.7%	39.2	Area = 6200 Sq. Ft	$6200 \times \frac{100}{10}$ = 62,000 Tons	0.75%
		118	63.5	0.8%	50.8			
		Average	119.5	0.75%	90.0			
Area ③	Elev. 5300	D.D. Hole 115	61.5'	0.9%	55.4	Area = 3200 Sq. Ft	$3200 \times \frac{100}{10}$ = 32000 Tons	1.24%
		118	31.5'	1.9%	59.8			
		Average	93.0	1.24%	115.2			
Area ④	Elev. 5668	Cave Adit Samples	43'	2.5%		Area = 12210 Sq. Ft.	$12210 \times \frac{100}{10}$ = 122100 Tons	1.73%
			105'	1.3				
			35'	1.4				
		Average		1.73%				
Area ⑤	Elev. 5500	D.D. Hole 38	39'	1.4%	54.6	Area = 2980 Sq. Ft.	$2980 \times \frac{100}{10}$ = 29800 Tons	1.12%
		39	50'	0.9%	45.0			
			89	1.12%	99.6			
Total							340900	1.23%

Centre Zone

Area ⑥	Elev 5600	Centre Adit	47'	1.4%		Area = 500 Sq. Ft 2000	$500 \times \frac{50}{10}$ = 2500 Tons	1.40%
Area ⑦	Elev 5300	D.D. Hole 40	10'	1.4%	14.0	Area = 3580 Sq. Ft	$3580 \times \frac{100}{10}$ = 35800 Tons	1.36%
		37	33'	1.35%	44.5			
			43	1.36%	58.5			
Total							38300 Tons	1.37%

New Zone

Area ⑧	Elev. 5500	D.D. Hole 31	11'	0.9%	9.90	Area = 760 Sq. Ft 2000	$760 \times \frac{100}{10}$ = 7600 Tons	0.8%
		33	11	0.7%	7.70			
			22	0.8%	17.60			
Area ⑨	Elev. 5130 50	D.D. Hole 40	13'	0.8	10.4	Area = 4700 Sq. Ft	$4700 \times \frac{100}{10}$ = 47000 Tons	1.27%
		74	7	0.3	2.1			
		37	28	2.1	58.8			
		91	17.5	.54	9.5			
		38	18	1.4	25.2			
			83.5	1.27	106.0			
Total	470						54600 Tons	1.20%

200

$\frac{20000 \times 500}{50 \quad 10}$ 100000

100000 @ 1%

$\frac{2000 \times 370}{10}$ 74,000

HILL, STARCK & ASSOCIATES
CONSULTING ENGINEERS
VANCOUVER, BRITISH COLUMBIA

Cowichan Copper Co., Ltd.
Sunro Mine - Cave, Centre & New Zones
Ore Reserve Calculations

DATE June 28 '63	SCALE —
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